Strategic Logistics Management

Course 5 — Advanced Supply Chain and Strategic Logistics Management · Module 15 · Teaching content for the Theory Builder

Review copy. Each lesson lists its blocks in entry order: enter each block's heading and body into the Theory Builder; enter practice questions as Lesson Checks; enter the module assessment via Configure assessment. Grey appendix sections are audit notes, not for entry.

Lesson 15.1 — Logistics Strategy and Service-Provider Selection

Summary

This lesson examines how organisations develop logistics strategies aligned with business objectives and customer requirements. Learners will explore strategic decisions involving transportation networks, outsourcing, service-provider selection, cost management, and service quality. The lesson also introduces structured supplier evaluation and explains how organisations balance operational capabilities, financial considerations, reliability, and long-term business needs.

Learning objectives

- Define logistics strategy and explain its relationship with organisational objectives.
- Identify major strategic decisions affecting logistics operations.
- Explain how customer requirements influence logistics strategy.
- Distinguish between in-house logistics and outsourced logistics services.
- Identify criteria used to evaluate logistics service providers.
- Explain the importance of cost, reliability, capacity, compliance, and service quality.
- Describe the basic stages of service-provider selection.
- Explain how supplier performance can be monitored and reviewed.
- Recognise the risks associated with excessive dependence on a single service provider.
- Develop a basic service-provider evaluation framework.

Key terms

Logistics Strategy — A long-term approach to organising logistics resources and activities to support defined business objectives.

Strategic Planning — The process of establishing long-term objectives and determining actions and resources required to achieve them.

Service-Provider Selection — The process of evaluating and choosing external organisations to deliver specified services.

Logistics Outsourcing — The use of external service providers to perform logistics functions.

Third-Party Logistics (3PL) — The provision of outsourced logistics services by an external organisation.

Fourth-Party Logistics (4PL) — A logistics management arrangement in which a provider coordinates or integrates multiple logistics resources and service providers on behalf of a customer.

Request for Quotation (RFQ) — A formal request inviting suppliers to submit prices and commercial terms for specified requirements.

Request for Proposal (RFP) — A formal request inviting suppliers to propose solutions, service approaches, and commercial arrangements.

Supplier Evaluation — The systematic assessment of a supplier's capabilities, performance, costs, and suitability.

Total Cost of Ownership (TCO) — The total cost associated with acquiring, using, managing, and maintaining a product or service over a defined period.

Supplier Diversification — The use of multiple suppliers or service providers to reduce dependence and improve flexibility.

Strategic Alignment — The consistency between operational decisions and broader organisational objectives.

Vendor Performance — The measured effectiveness of a supplier or service provider in meeting agreed requirements.

Teaching content

KEY CONCEPT

What a logistics strategy is

Every operational detail in this programme — the delivery standards of the lesson on the Seven Rights of Logistics, the service-level targets of the lesson on Inventory Optimisation and Service Levels, the cost components of the lesson on Components of Total Logistics Cost — was decided somewhere, by someone, on purpose. Strategy is the level where those decisions are made.

A logistics strategy is a long-term approach to organising logistics resources and activities to support defined business objectives. In plainer terms, it is the deliberate answer to three questions. What service do we promise our customers? Through what structure — our own people and assets, contracted providers, or a blend — will we deliver that promise? And at what cost, so that the promise remains profitable?

The word strategy places the activity inside strategic planning, the process of establishing long-term objectives and determining the actions and resources required to achieve them. Logistics strategy is that planning applied to the movement and storage of goods: the business decides where it is going, and the strategy decides what logistics must be to get it there.

The test of a good strategy is strategic alignment — the consistency between operational decisions and broader organisational objectives. A business whose promise is speed cannot run a structure built to minimise cost per load above all else; a bulk trader whose customers accept weekly deliveries should not pay for a same-day machine. The lesson on the Seven Rights of Logistics listed what every delivery must get right; the strategy decides which of those rights this particular business will guarantee most firmly, and what it will pay to guarantee them.

A logistics strategy is therefore not a slogan on a shelf. It is a position, visible in concrete commitments: which vehicles are owned, where warehouses stand, which providers hold contracts, and which performance measures those contracts carry. This lesson builds that position — first its decisions, then the discipline of selecting the providers who will carry much of it.

KEY CONCEPT

Customer requirements shape the strategy

Logistics strategy starts outside the logistics department. It starts with what the customer has been promised, because the promise dictates the structure.

Consider two businesses. A retailer promising same-day delivery of consumer orders needs dense local coverage, small vehicles moving constantly, tight order cut-off times, and systems that show every order's status hour by hour. A trader selling cement in bulk, delivered in full loads to construction sites with agreed lead times, needs a completely different machine: large vehicles, port or plant loading capacity, and disciplined scheduling of fewer, bigger deliveries. Neither structure is better: each is right for its promise, and absurd for the other.

Customer requirements reach the strategy through several channels: the speed customers expect, the sizes and frequency of their orders, the condition the goods must arrive in (frozen fish, medicines and machine parts demand very different care), the geography that must be covered, and the information customers expect — proof of delivery, shipment tracking, accurate documents.

The lesson on Inventory Optimisation and Service Levels showed how service-level targets translate customer expectations into measurable commitments, such as dependable product availability. The logistics strategy does the same for the whole operation: it takes the service promise, quantifies it, and then designs the structure — network, assets, providers — that can hit those targets at a cost the business can afford.

Two disciplines follow. First, the promise must drive the structure, not the other way round: a strategy copied from another company fails quietly because the promises differ. Second, the

promise must be affordable: the strongest strategies align promise, structure and cost so the business can keep the promise profitably, year after year.

KEY CONCEPT

The major strategic decisions in logistics

A logistics strategy becomes real through a small set of large decisions, taken together. The principal ones are these.

1. Which activities to own and which to outsource

Transport, warehousing, forwarding, fleet maintenance — for each, the business asks whether to perform it with its own people and assets or buy it from the market. This make-or-buy decision carries the whole weight of outsourcing, and deserves its own treatment below.

2. Centralise or decentralise the network

How many warehouses and distribution points, and where? More, deeper networks place stock closer to customers and shorten final deliveries, but add facilities, inventory and fixed cost. A centralised network does the reverse: cheaper to run, but goods travel further to reach the customer. The right answer depends on the service promise — the same-day retailer and the bulk trader of the previous section would centralise at opposite ends.

3. Asset-heavy or asset-light

Should the business own vehicles and warehouses, or rent access to other people's capacity? Owning brings control and committed capacity, but locks capital into assets that must be kept busy. Asset-light operation keeps capital free and flexes with demand, but depends on the market's willingness to sell that capacity when it is needed.

4. Single provider or several

Concentrating volumes with one provider buys scale, leverage and simplicity; spreading them across several buys resilience and choice. This trade-off — and the risks of concentrating too far — is examined at the end of the lesson.

5. Where to position cost and service quality

Cost management and service quality are strategic positions, not accidents: the strategy decides which service elements the business competes on and which costs it manages hardest, because maximising both everywhere produces an operation that is expensive without being excellent.

These decisions interlock: an outsourcing choice changes the network options; a network choice changes what providers are needed. Together they form the structure half of the strategy. The first — own or outsource — shapes all the others, so it comes next.

KEY CONCEPT

In-house or outsourced: the make-or-buy logic

Logistics outsourcing is the use of external service providers to perform logistics functions. Its most common vehicle is third-party logistics — 3PL — the provision of outsourced logistics services such as transport, warehousing and forwarding by an external organisation, introduced in the lesson on Types of Logistics: Inbound, Outbound, Reverse, and Third-Party. Above the 3PL sits fourth-party logistics — 4PL — an arrangement in which a provider coordinates or integrates multiple logistics resources and service providers on behalf of the customer, typically owning few or no transport and warehouse assets: a 3PL operates logistics for you, while a 4PL manages the whole network, including the 3PLs.

Whether to outsource at all is the make-or-buy decision, and it should be walked through four factors rather than decided by habit.

1. Capability
Can we perform this activity as well as a specialist would? Providers move many customers' goods through shared networks, spreading fixed costs, employing trained specialists and running professional systems; a manufacturer's core excellence is manufacturing, and transport is the provider's core.

2. Volume
Is our volume large and steady enough to keep owned resources busy? Own vehicles and warehouses earn nothing when they stand idle, and idle capacity is paid for either way. Seasonal peaks and troughs argue for buying capacity only when it is needed.

3. Control
How much direct control does this activity genuinely need? The more strategic or sensitive the flow — customer-critical deliveries, tightly regulated goods — the stronger the case for keeping it close, or at least for holding control through strict performance measures rather than through ownership.

4. Capital
What would ownership tie up? Vehicles, warehouses and equipment consume capital that could fund the core business, and bring maintenance, insurance and replacement obligations; outsourcing converts that commitment into a purchasable service.

Rarely does one factor decide alone: the discipline lies in answering all four honestly for each activity, then respecting the answer — as the first worked example of this lesson does, end to end.

KEY CONCEPT

What to outsource, what to keep, and the hidden costs

The make-or-buy factors compress into a working rule. Outsource what the market can do better or cheaper at scale, and what is not your strategic core. Keep what differentiates you from competitors, and what you cannot safely surrender control of. A cocoa exporter's differentiating capability is its buying and quality system, not its truck fleet; a pharmacy chain's differentiating capability may include the cold-chain handling its medicines depend on.

But outsourcing is not free, and a strategy that counts only the invoice has missed three hidden costs.

1. Coordination
Every outsourced activity adds a party whose work must be specified, scheduled and checked. The lesson on Coordinating Carriers, Agents, and Regulatory Authorities showed how much communication a single international movement already demands; each new provider multiplies the contracts, data exchanges and meetings that hold the operation together.

2. Reduced visibility
Work done in-house can be watched directly. Work done by a provider is seen through reports and systems — or not at all. Outsourcing without visibility arrangements is outsourcing blind.

3. Dependency
Depend on a provider long enough and the internal skills to specify, judge and replace that provider quietly fade. The knowledge needed to run the activity migrates out of the business, and with it the power to leave the relationship.

Most businesses therefore live in the hybrid middle: keeping in-house the activities that are core, sensitive or short-notice, and contracting out the predictable, scale-dependent work. The same blending applies to how they buy capacity — core lanes under contract rates for committed volumes, with peaks covered by spot purchases at current market prices. Ownership versus outsourcing is not a yes-or-no vote taken once. It is a position, maintained activity by activity.

KEY CONCEPT

Specification first, then the stages of selection

Once the strategy says "buy", the discipline of service-provider selection begins — the process of evaluating and choosing external organisations to deliver specified services. It rests on one uncomfortable truth: you cannot evaluate what you have not specified.

Specification means writing down, before any provider is contacted, what is actually being bought: the scope of services (which activities, which stages); the lanes and routes (origins, destinations, frequencies); the volumes (now and expected over the contract term); the service promise (delivery windows, condition requirements such as cold chain, information needs); and

the commercial horizon (contract length, expected growth). A weak specification lets every bidder price a different imaginary service, produces bids that cannot be compared, and saves its surprises for the first month of operation, when they are most expensive.

With the specification written, selection follows recognisable stages.

1. Specify needs
The scope, volumes, service promise and evaluation priorities, in writing.

2. Scan the market and build a long list
Identify every plausible provider from industry contacts, associations and existing relationships — a broad first list, with judgement applied only for basic plausibility.

3. Short-list and invite proposals
Screen the long list against minimum requirements — capability, coverage, licences — and issue the surviving short list a formal request. A request for quotation, or RFQ, invites suppliers to submit prices and commercial terms for specified requirements, and suits standardised, fully specified services. A request for proposal, or RFP, invites suppliers to propose solutions, service approaches and commercial arrangements, and suits complex services where the how matters as much as the price. Either way, every bidder receives the same scope and the same questions — the quotation discipline taught in the lesson on Preparing and Evaluating Logistics Quotations, applied at contract scale.

4. Evaluate the proposals
Scored against criteria agreed in advance — the subject of the next section.

5. Verify what the paper claims
Reference checks and site visits, before commitment rather than after.

6. Contract, implement, and monitor
The award is not the finish: the measures and review rhythm that follow are where selection succeeds or quietly fails.

KEY CONCEPT

Evaluation criteria: price comes last, not first

Supplier evaluation is the systematic assessment of a supplier's capabilities, performance, costs and suitability. Its quality depends almost entirely on the criteria chosen and the order they are considered in — and five areas the syllabus names outright, cost, reliability, capacity, compliance and service quality, each appear below. The professional criterion set covers seven areas.

1. Capability and coverage
Does the provider actually perform the services in the specification — the right vehicles, warehouses, handling equipment and skilled people — across the required geography? Include

capacity: enough equipment and staff not only for today's volumes but for seasonal peaks and expected growth. A provider that cannot scale will serve the first year well and disappoint the third.

2. Reliability and track record
What has this provider actually delivered before, for customers like us, on lanes like ours? References, contract history and performance evidence matter more than promises: past behaviour on comparable work is the best predictor of future behaviour.

3. Systems and visibility
Can the provider show where shipments are, exchange data with our systems, and produce the reports the contract will require? The lessons on Transport and Warehouse Management Systems and on electronic data exchange established what good systems look like; a provider without them drags the customer's own visibility down to the provider's level.

4. Compliance and safety record
Licences, insurance, customs compliance, dangerous-goods competence where relevant, and safety history. Compliance failures do not stay with the provider — they land on the customer's shipment, as this lesson's important notes emphasise.

5. Financial health
A provider that fails commercially mid-contract halts the customer's operation with it. Solvency and staying power are operational criteria, not merely financial ones.

6. Cultural fit
Responsiveness, honesty when things go wrong, willingness to solve problems jointly. Contracts are operated by people, and mismatched ways of working erode even well-drafted agreements.

7. Price
Last, deliberately: price is meaningful only once the service being priced is genuinely comparable, which is why the specification came first. The cautionary tale later in this lesson shows what forgetting this costs.

KEY CONCEPT

The weighted framework, reference checks, and site visits

Criteria alone still leave a problem: different evaluators weigh reliability, price and compliance differently in their heads, and the loudest voice wins. The remedy is a written evaluation framework — a scored comparison agreed before any proposal is opened.

The framework has three parts. First, criteria drawn from the specification, each given a weight reflecting the service promise: a cold-chain pharmaceutical contract weights compliance and reliability far above price; a bulk construction-materials lane might weight capacity and cost more heavily. Second, a common scoring scale applied identically to every bidder on every criterion. Third, a documented calculation — each score multiplied by its weight, summed to a

total — together with the reasoning behind the scores. The documentation matters as much as the arithmetic: it keeps the evaluation honest under pressure, lets the loser be told fairly why they lost, and protects the business if the decision is later questioned.

Two verification steps complete the discipline, and neither can be done from a desk.

Reference checks mean contacting the short-listed provider's existing customers — ideally of similar size and lanes — and asking specific questions: what was actually delivered against the promise, what happened when a delivery failed, how the provider behaved when things went wrong. Vague praise is worth little; a concrete account of failure and recovery is worth a great deal.

Site visits mean seeing the operation as it really is: the condition and age of the vehicles and equipment, the state of the warehouse, whether the tracking system demonstrated in the pitch is the system actually used in daily work, and how the management team answers unwelcome questions. Proposals are written by salespeople; sites are run by operations.

Framework, references and visit together convert selection from an impression into an evidence-based decision — exactly what a strategic commitment of several years deserves.

KEY CONCEPT

Total cost of ownership

The final evaluation discipline corrects the most common error in provider selection: comparing quotations as if the quoted price were the cost.

Total cost of ownership, or TCO, is the total cost associated with acquiring, using, managing and maintaining a product or service over a defined period. Applied to a logistics contract, the quoted rate is only the visible layer. Beneath it sit the costs the contract generates in operation: the coordination effort of managing the provider; the cost of claims when goods are damaged; the cost of expediting when deliveries miss their windows; the compensation and lost custom when the service fails the end customer; and eventually the cost of switching away.

This is the total-cost thinking of the lesson on Components of Total Logistics Cost — transport, warehousing, inventory holding, handling, administration and insurance seen together, direct and indirect costs both counted — turned onto a single purchasing decision. A provider whose price sits somewhat above the lowest bid but whose reliability prevents claims, expediting and lost sales is frequently the cheaper choice over the contract term, because the avoided failures are real money even though they never appear on a rate card.

TCO therefore changes how bids should be read: estimate, for each bidder, not only what the rate will cost but what the bidder's performance record is likely to cost in failures and management attention, over the full contract period. The decision rule is simple to state: compare bids on total cost over the term, not on the quotation. The third worked example of

this lesson shows, in the negative, what happens when a business forgets it — an award made on price alone that repaid the saving several times over in claims and missed deliveries.

KEY CONCEPT

From selection to partnership: monitoring and governance

Selecting a provider creates a relationship; the strategy must then say what kind of relationship it intends to run.

The instrument of that relationship is the service-level agreement, or SLA — an agreement that states what service is bought, how it is measured, what happens when it is missed, and how either side exits. Its standard components form a checklist every logistics manager should carry: the scope of services; the KPIs with agreed targets — drawn from the measures of the lesson on Logistics Performance Indicators and KPIs, such as on-time in-full delivery per lane or damage rates; the measurement and reporting arrangements; the remedies and service credits that apply when targets are missed; the governance and review cycle; and the exit and transition terms. The next lesson, Contract Management and Service-Level Agreements, treats each component in depth; the point here is that the SLA is where the strategy's service promise becomes enforceable.

Monitoring turns those measures into vendor performance — the measured effectiveness of a provider in meeting agreed requirements. A working rhythm has three layers: regular performance reports against the KPIs; scheduled reviews — commonly quarterly — where trends, recurring misses and root causes are discussed; and joint improvement actions, so both parties work from the same data on making the next period better. This rhythm is also the early-warning system for everything this lesson has warned about: failing capability, growing dependency, eroding financial health.

The mindset must be chosen per relationship. A transactional mindset — buy, receive, pay — suits spot-market commodity purchases, where switching costs nothing and loyalty earns nothing. A partnership mindset suits the strategic lanes and multi-year contracts at the centre of the strategy: it shares information and forecasts, plans jointly, and gives both sides reason to invest in the relationship. Treating strategic partners as disposable suppliers forfeits the commitment the strategy paid for; treating spot purchases as partnerships overpays for them.

KEY CONCEPT

Dependence and diversification

The last strategic decision — single provider or several — deserves its own examination, because its risks arrive late and all at once.

Excessive dependence on a single service provider is a quiet concentration of risk. Operationally, the provider's failure — insolvency, capacity shortage, a damaged fleet, a lost licence — becomes the customer's failure overnight, with no ready alternative. Commercially, dependence reverses the bargaining position: the provider who knows the customer has nowhere else to go

is the provider who can raise rates and relax effort. Strategically, dependence deepens itself: each year of single-provider operation transfers more knowledge out of the customer's business, until even evaluating an alternative becomes difficult.

The strategic answer is supplier diversification — the use of multiple suppliers or service providers to reduce dependence and improve flexibility. It takes several forms: splitting critical lanes between two qualified providers, so each keeps the other honest and both together carry the volume; keeping a second provider qualified and lightly used, so that shifting volume in a crisis is a decision rather than a project; and blending contract rates for core volumes with spot purchases for peaks, so no single provider holds all the capacity.

Diversification has a price, and honesty requires naming it. Volume split between providers is volume concentrated with neither: rates rise, committed capacity shrinks, and the scale that made the relationship attractive thins out. The strategy therefore balances concentration — for leverage, scale and depth of partnership where the flows are stable and the provider proven — against diversification — where the flows are critical and the cost of failure high. How much dependence is acceptable is itself a strategy decision, reviewed as the business and the market change.

WORKED EXAMPLE

Make or buy: the fleet decision at a Tema juice manufacturer

A manufacturer of packaged fruit juice in Tema distributes to wholesalers in Accra, Kumasi, Takoradi and Cape Coast. Its delivery volumes rise sharply before festive seasons and fall away for much of the rest of the year. Its own trucks are ageing, and the transport manager has proposed replacing them with a new fleet. The board applies the make-or-buy logic instead. The scenario and its figures are illustrative for teaching purposes.

Step 1: Capability

The company's own drivers know the routes and the returnable-crate handling, but the fleet is maintained reactively, delivery planning lives in one person's head, and no tracking exists. Specialist distributors in the region run scheduled maintenance, trained drivers and shipment tracking as standard products. On capability alone, the market would perform the core line-haul better.

Step 2: Volume

The analysis shows the fleet fully stretched for roughly a quarter of the year and substantially idle for the rest — yet drivers are paid and vehicles depreciate all year. Ownership would buy capacity for the peak and pay for it in the trough. Volume argues for buying transport as a service.

Step 3: Control

The goods are robust, customer deliveries are to wholesale depots with agreed windows, and nothing in the flow requires licences or special handling. What the customers actually need is

reliability, not ownership — and reliability can be held through a contract with delivery-window measures and remedies rather than through painted company names on doors.

Step 4: Capital
Quotations show a new fleet plus a workshop, spare-parts stock and driver payroll would absorb capital the production side has been requesting for two expansion projects. Trucks would be the company's largest single asset, serving its least differentiating activity.

Step 5: The decision
The company contracts a regional 3PL for line-haul under agreed rates for committed base volumes, with peak capacity bought additionally in high season, and keeps two small company vehicles for city top-ups and short-notice orders. Control is exercised through a service-level agreement carrying delivery and condition measures, reviewed each quarter.

What does this example demonstrate?
The four factors — capability, volume, control, capital — walked honestly to a position rather than a reflex. Key lesson: ownership is only one form of control; in this case the contract delivered more reliability than the fleet ever had, released the capital, and stopped paying for idle trucks.

WORKED EXAMPLE

Selection done right: a pharmaceutical distributor awards a regional contract

A pharmaceutical distributor based in Accra supplies wholesalers in Kumasi, Takoradi and Tamale with temperature-sensitive medicines and decides to outsource regional distribution. The scenario, weights and scores in this example are illustrative for teaching purposes.

Step 1: Specify
Before contacting anyone, the distributor writes the specification: collect palletised stock from the Accra warehouse under cold-chain conditions; deliver next-day to Kumasi and Takoradi wholesalers and within two days to Tamale; maintain temperature records for every consignment; provide electronic proof of delivery the same day; handle current volumes with headroom for growth. The specification states the priorities outright: reliability and compliance first, price later.

Step 2: Scan and short-list
A market scan through industry contacts and professional associations produces a long list of eight; screening against minimum requirements — cold-chain capability, national coverage, valid licences and insurance — shortens it to five.

Step 3: Invite proposals
All five receive the same specification document and the same structured questions — the same scope for every bidder, so answers are comparable. Because the service is complex and multi-year, this is a request for proposal, inviting each provider to describe its service approach as well as its price.

Step 4: Evaluate with weights
The panel had agreed criteria and weights before opening any proposal: service capability and cold-chain competence twenty-five per cent, reliability and track record twenty per cent, compliance and insurance twenty per cent, systems and visibility fifteen per cent, cultural fit ten per cent, price ten per cent. Each bidder is scored on every criterion. The weighted totals place first overall a provider ranking only second on price — not the cheapest bid, but the strongest on reliability, compliance and systems.

Step 5: Verify
Reference checks call three of the winner's customers of similar profile, asking specifically about temperature excursions, missed deliveries and behaviour after failures; the answers confirm the record. A site visit inspects the depot — refrigerated vehicles and maintenance records, cold rooms, and the monitoring system working in live operation, not in slides.

Step 6: Contract and monitor
The contract annexes the specification and a service-level agreement carrying the measures of the lesson on Logistics Performance Indicators and KPIs — on-time in-full delivery per lane, delivery cycle time, temperature excursion rate, proof-of-delivery timeliness — with monthly reporting, remedies for misses, quarterly reviews and exit terms. The next lesson, Contract Management and Service-Level Agreements, examines that agreement in depth.

What does this example demonstrate?
The full selection discipline: specify, scan, short-list, invite, score with agreed weights, verify, contract with measurable commitments. Key lesson: because the specification came first, every bid was comparable, the evaluation was defensible, and the contract inherited the promise instead of inventing it afterwards.

WORKED EXAMPLE

The price-first award: a cautionary tale from Mumbai

A furniture retailer in Mumbai distributes imported flat-pack furniture from a central warehouse to its stores across the surrounding region. When its distribution contract comes up for renewal, the retailer invites proposals from five providers. The scenario is illustrative for teaching purposes.

Step 1: The award
There is no written service specification — only a rough description of the work and a request for prices. The bids come back structured differently and describing different services, so they cannot be compared on anything but number. The award goes to the lowest quote, which sits noticeably below the rest of the field. The evaluation takes an afternoon.

Step 2: The first months
Problems arrive on a predictable schedule, each tracing back to the service the low price never included. Vehicles are older than promised, so breakdowns push deliveries past agreed store

windows. Handlers are casual and untrained, so furniture arrives scratched and with corner damage; claims from the stores mount. There is no shipment tracking, so store managers spend their mornings telephoning the depot for whereabouts and their afternoons arranging premium transport for the shortfalls.

Step 3: The full bill
At the quarterly review the retailer finally assembles the whole picture. Against the saving made on the quotation, it sets: the claims paid for damaged furniture; the premium transport bought to patch missed deliveries; the compensation owed to stores left understocked in trading weeks; the management hours consumed by chasing; and the customers who bought elsewhere when shelves were bare. Measured this way, the cheapest contract costs more than the next-best bid would have — the price gap was an illusion created by comparing quotations instead of services.

Step 4: The re-award
The contract is retendered properly: a written specification, the same scope to all bidders, evaluation on weighted criteria, and bids compared on total cost of ownership over the contract term — the rate plus a realistic estimate of each bidder's failure costs and management burden. The new winner is a mid-priced provider with strong references, and the failure costs it avoids repay its price premium within the first year.

What does this example demonstrate?
The difference between price and cost, and the failure chain that runs from a missing specification through an unscorable comparison to an award on the only number available. Key lesson: the cheapest bid is cheap only if the service arrives; every avoided claim, expedite and lost sale is money the rate card never mentioned.

IMPORTANT NOTE

Outsourcing transfers the task, not the accountability

The most dangerous sentence in outsourcing is "that is the provider's problem." The task is transferred; the accountability is not.

When a manufacturer's contracted carrier delivers late, the retailer does not ask about the carrier — it stops ordering from the manufacturer. When a distributor's provider breaks the cold chain, the pharmacy blames the distributor, and regulators may do the same. The customer experience belongs, always, to the business that made the promise. This is the same principle the lesson on the Role and Responsibilities of Freight Forwarders established from the other side: a forwarder acts on the customer's behalf, yet the shipment remains the customer's — and the same holds when the business itself is the customer of a 3PL.

Accountability has practical consequences. A provider must be managed actively, not merely engaged: specification, KPIs, reporting and reviews exist precisely because the business remains answerable for work it no longer performs. Enough internal knowledge must be retained to

specify the service, judge the performance and, when necessary, leave the relationship — the antidote to the dependency described in this lesson. And failures should be treated as the business's own failures to fix: the first response to a provider's miss is not to quote the contract at them but to establish what happened, hold the provider to the agreed remedies, and repair the customer relationship, which is the only one the end customer can see.

Outsourcing, done well, is delegation with control. Done carelessly, it is abdication with paperwork.

IMPORTANT NOTE

A contract without measurable KPIs is an aspiration

Provider contracts frequently contain sentences like "the provider will deliver in a timely and professional manner." Such sentences are aspirations, not commitments: no one can say what "timely" means, so no one can ever be proved to have missed it. Where nothing is measurable, nothing is enforceable — no remedy can be invoked, no trend can be tracked, no improvement can be demonstrated, and disputes decay into whose-word-stands arguments.

The remedy is to give every promise three companions: a number, a measurement method and a consequence. "Next-day delivery" becomes on-time in-full delivery per lane, counted as the lesson on Logistics Performance Indicators and KPIs prescribes — an order counts only if it arrives both on time and complete — measured from a defined data source, reported monthly, with an agreed remedy when the target is missed. Every service element in the specification should pass the same test: if it cannot be measured, either define the measurement or admit it is a preference, not a commitment.

Two cautions keep measurement honest. First, few measures well: a contract carrying dozens of KPIs is as blind as one carrying none, because nobody manages what nobody reads. Second, people manage what is measured, so a badly chosen measure gets gamed — count only on-time deliveries and depots will be emptied of stock to make the window. Measures must be chosen to reflect the promise the customer actually bought.

The next lesson, Contract Management and Service-Level Agreements, builds the full architecture; the working rule to carry into it: a contract that cannot be measured cannot be managed, and a promise without a number is a hope.

IMPORTANT NOTE

Strategy is a position, not a portrait

A logistics strategy is a position taken about service, structure and cost — and positions depend on conditions. When the conditions move and the position does not, yesterday's strategy becomes today's habit, defended by habit rather than by results.

Three kinds of change should trigger review. The business changes: new products, new markets, new channels — a strategy built for wholesale distribution creaks when the company opens an

online shop promising direct delivery. The volumes change: growth can break a centralised network that once served brilliantly, while decline strands owned assets and starves contracted volumes below the levels the rates were built on. The market changes: providers merge or fail, capacity tightens or floods, and rate levels move — including after the kind of freight-market disruption the industry has repeatedly seen in recent years. Each shift alters what is available, what it costs, and how much dependence on any single provider is safe.

Review should therefore run on two clocks. The scheduled clock — the governance reviews of the provider contracts, and a periodic look at the whole strategy — catches slow drift. The trigger clock — an explicit decision to re-examine the strategy whenever the business, the volumes or the market shift materially — catches the sudden moves before the operation spends a year compensating for a position that no longer exists.

None of this means strategy changes casually: a position worth taking is worth holding while its conditions last — held conditionally, and examined. The professional test is simple to state and demanding to apply: could we currently explain, with present facts, why our logistics is structured the way it is? When the honest answer becomes "because it always has been", the review is overdue.

Practice questions (Lesson Checks)

Practice Question 1

A producer of shea butter in Tamale ships palletised consignments to distributors in Accra and Tema. Volumes rise sharply during harvest season and fall away for much of the rest of the year. The production manager proposes buying a fleet of trucks, arguing that ownership would give the company control of its deliveries.

Which option best describes the sound strategic analysis of this make-or-buy decision?

1. The decision should weigh capability, volume, control and capital: seasonal volumes would leave owned vehicles idle for much of the year while capital sits locked in them, so contracting transport — with reliability held in place through agreed performance measures — is likely the sounder position. ✓

2. The company should buy the fleet, because owning vehicles is always cheaper in the long run than paying a service provider for the same journeys.

3. The company should outsource all transport and stop monitoring deliveries afterwards, because a contract transfers accountability for delivery performance to the provider.

4. The company should decide purely by choosing whichever transport provider submits the lowest quotation, since price is the only factor that matters.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

The make-or-buy decision weighs four factors. Capability: specialists run transport as their core business, with systems and maintenance an individual manufacturer struggles to match. Volume: this producer's demand is strongly seasonal, and owned vehicles earn nothing while standing idle for much of the year — the business would be buying peak capacity and paying for it all year. Control: what the customers need is reliability, which a contract can secure through KPIs and remedies; ownership is only one form of control. Capital: a fleet would absorb capital that serves the company's core business better. All four point the same way, which is what makes the position sound rather than merely defensible.

Answer 2 ignores utilisation and capital: ownership is cheaper only when volume keeps the assets busy. Answer 3 repeats the misconception that outsourcing transfers accountability — the task moves, but the customer still experiences the failure, so monitoring must continue. Answer 4 is the price-first error this lesson dismantles.

The takeaway: own or outsource is a four-factor analysis, not a reflex — and a contract can hold control better than a parked truck.

Practice Question 2

A distributor in Accra is selecting a provider for regional distribution. Two proposals remain. The lowest-priced bid comes from a provider with a weak delivery record, ageing vehicles and no shipment tracking. A higher bid comes from a provider with strong references from similar customers, complete compliance documentation and live tracking.

Which statement best describes sound provider selection in this situation?

1. Price is the first criterion in provider selection, so the lowest bid should win unless a competitor matches it exactly.

2. Selection should weigh all the specified criteria — reliability, capacity, compliance, systems and service quality — and compare the bids on total cost of ownership, because service failures carry costs that can exceed the price gap. ✓

3. The largest and best-known provider available should be appointed, because the size of a company guarantees its performance.

4. The contract should be split equally between the two providers so that neither is disappointed and no difficult choice has to be made.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Provider selection evaluates capability and coverage, reliability and track record, systems and visibility, compliance and safety, financial health and cultural fit — with price considered last, after the service being priced is genuinely comparable. Here the bids describe very different services: the cheap bid buys an operation whose record predicts claims, missed windows and management time, and total cost of ownership counts those failure costs alongside the rate, as this lesson's Mumbai example showed. The dearer bid's references, compliance and tracking are precisely the criteria that prevent such failures.

Answer 1 is the price-first fallacy the lesson warns against: comparing quotations as if they priced the same service, when the specification and the criteria say otherwise. Answer 3 substitutes size for fit — large providers also fail when capability, coverage and cultural fit are wrong. Answer 4 misuses diversification: splitting volume is a deliberate risk decision with its own cost, not a way to avoid choosing between a sound bid and a weak one.

The takeaway: evaluate against the specification, price last, and decide on total cost over the contract term.

Practice Question 3

A manufacturer depends on a single logistics provider for all of its transport and warehousing. Concerned about this concentration, the board asks management which actions would genuinely reduce the risks of excessive dependence on one provider.

Which option is NOT a sound response to excessive dependence on a single service provider?

1. Qualifying a second provider on the critical lanes, so that volume can be shifted deliberately if the incumbent fails.

2. Blending contracted rates for core volumes with spot purchases for peaks, so that no single provider holds all the capacity the business needs.

3. Cancelling the incumbent's contract immediately and moving all volume to a new provider found at short notice, without evaluation, specification or transition arrangements. ✓

4. Holding regular performance reviews with the incumbent, so that warning signs of failure surface while options to respond still exist.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

Dependence on one provider concentrates risk: the provider's failure becomes the customer's failure overnight, the bargaining position reverses, and the internal knowledge to switch fades. The sound responses reduce that exposure with discipline — qualifying a second provider before a crisis makes switching a decision rather than a project; blending contract and spot cover so no one provider holds all the capacity; and using the governance rhythm of reviews and KPIs as the early-warning system that surfaces problems in time. Answer 3 does the opposite of all three: it replaces a managed position with an unmanaged leap, repeating the entire selection process in its worst form — no specification, no evaluation, no references, no transition — and would likely deepen dependence on the new provider within months.

Dependence is reduced by prepared alternatives and early warning, not by panic. Diversification is a strategy, and strategies are specified and evaluated like everything else in this lesson.

The takeaway: the opposite of excessive dependence is not haste — it is qualified alternatives, measurable performance, and a review rhythm that sees trouble coming.

Fact-check appendix (audit notes — not for entry)

Definitions of logistics strategy, strategic planning, service-provider selection, logistics outsourcing, third-party logistics (3PL), fourth-party logistics (4PL), request for quotation (RFQ), request for proposal (RFP), supplier evaluation, total cost of ownership (TCO), supplier diversification, strategic alignment and vendor performance follow the lesson brief key terms. — source: lesson brief 15.1 key terms (curriculum database).

3PL described as the outsourcing of logistics functions — transport, warehousing, forwarding, pick-and-pack — to an external provider acting between seller and buyer. — source: research sheet logistics-scm-frameworks.md, section 1 (Third-party logistics).

4PL described as an asset-light integrator or lead logistics provider that coordinates or integrates multiple 3PLs and technology on behalf of a customer, typically owning few or no assets; no year of origin taught and no vendor named. — source: research sheet logistics-scm-frameworks.md, section 1; DO NOT CLAIM item 4 (no 4PL invention year; attribution omitted entirely per no-vendor-names rule) observed.

Make-vs-buy logic taught as the four factors capability, volume, control and capital, consistent with the research sheet's framing: weigh cost, capability, control, demand variability and strategic importance (cost folded into capital, demand variability into volume, strategic importance into control). — source: research sheet logistics-scm-frameworks.md, section 10 (Strategy and contracts).

Outsourcing rule of thumb "outsource when the market can do it better or cheaper and it is not your strategic core" — source: research sheet logistics-scm-frameworks.md, section 10 safe phrasings.

Hidden costs of outsourcing (coordination effort, reduced visibility, dependency) taught as general business knowledge consistent with the make-vs-buy framework. — source: general business knowledge, consistent with research sheet logistics-scm-frameworks.md, section 10.

3PL selection criteria taught as a checklist: service scope fit and coverage, capacity/scalability, track record and references, IT and visibility integration, compliance/insurance/safety, financial health, cultural fit, price last. — source: research sheet logistics-scm-frameworks.md, section 10 (3PL selection criteria; medium confidence, taught as checklist not canon), extended with financial health as general business knowledge.

SLA components taught as a checklist: scope of services; KPIs with targets; measurement and reporting cadence; remedies/service credits; governance and review cycle; exit/transition terms; summary phrasing "what service is

bought, how it is measured, what happens when it is missed, and how either side exits". — source: research sheet logistics-scm-frameworks.md, section 10 (SLA components; safe phrasing). Depth deliberately deferred to lesson 15.2 per WRITING_SPEC section 7 (anti-repetition).

Spot vs contract rates: spot is a one-off price at current market, contract is agreed rates for committed volume or period; shippers typically blend both — core lanes on contract, peaks on spot. — source: research sheet logistics-scm-frameworks.md, section 10.

OTIF defined as the percentage of orders delivered both on time and in full; strict OTIF fails an order that is late or short. — source: research sheet logistics-scm-frameworks.md, section 7 (Performance measurement).

Network design trade-off: more or deeper networks cut delivery time and outbound transport cost but add facility and inventory cost; centralised networks do the reverse. — source: research sheet logistics-scm-frameworks.md, section 8 (Planning, network design).

Total logistics cost as a total-cost view across transport, warehousing, inventory holding, handling, administration, insurance and distribution, with direct and indirect costs, referenced to the earlier lesson; the transport-versus-inventory total-cost trade-off principle underlies the TCO discussion. — source: research sheet logistics-scm-frameworks.md, sections 6 and 10; lesson brief 10.1 (Components of Total Logistics Cost).

KPI examples used in the SLA discussion (on-time in-full per lane, delivery/order cycle time, damage or claims ratio, inventory accuracy) are the standard measures of the performance-measurement lesson. — source: research sheet logistics-scm-frameworks.md, section 7; lesson brief 13.4 (Logistics Performance Indicators and KPIs).

Weighted-criteria evaluation with documented scoring, reference checks and site visits, and the RFP/RFQ discipline of issuing the same scope and structured questions to all bidders, are taught as standard procurement and provider-selection practice. — source: general business knowledge, consistent with research sheet logistics-scm-frameworks.md, section 10 and lesson brief 10.4 (Preparing and Evaluating Logistics Quotations).

Freight-market disruption referenced only qualitatively ("the kind of freight-market disruption the industry has repeatedly seen in recent years"), with no dates, percentages or magnitude claims. — source: research sheet logistics-scm-frameworks.md, section 11; DO NOT CLAIM items 6, 7 and 8 observed by omission.

No duty rates, fee amounts, statutory deadlines, free-time periods, currency amounts, vendor names or benchmark percentages are stated anywhere in the lesson; the evaluation weights and scores in the worked examples are explicitly labelled illustrative for teaching purposes. — source: WRITING_SPEC.md section 5 (fact-safety rules), rule 5.1.

Worked examples (Tema packaged-fruit-juice manufacturer; Accra pharmaceutical distributor with lanes to Kumasi, Takoradi and Tamale; Mumbai furniture retailer) are illustrative teaching constructions with no factual claims attached to real companies; locations and goods chosen from the spec's approved lists. — source: WRITING_SPEC.md section 4; teaching constructs.

Lesson 15.2 — Contract Management and Service-Level Agreements

Summary

This lesson examines the commercial and operational principles involved in managing logistics contracts and service-level agreements. Learners will explore contractual obligations, performance standards, pricing provisions, liability, reporting, dispute resolution, and contract review. The lesson also explains how clear agreements help manage expectations, reduce

commercial risks, and establish accountability between logistics customers and service providers.

Learning objectives

- Define contract management and service-level agreements.
- Explain the purpose of contracts in logistics service relationships.
- Identify essential provisions commonly included in logistics contracts.
- Distinguish between contractual obligations and operational performance targets.
- Explain how service-level agreements establish measurable expectations.
- Identify common logistics service-level indicators.
- Describe the importance of liability, insurance, and dispute-resolution provisions.
- Explain how contract performance is monitored and reviewed.
- Recognise the consequences of unclear or incomplete contractual terms.
- Evaluate a basic logistics service agreement against specified operational requirements.

Key terms

Contract Management — The administration and monitoring of contractual obligations throughout the life of an agreement.

Service-Level Agreement (SLA) — An agreement defining specified service expectations, performance measures, and responsibilities.

Contractual Obligation — A duty or responsibility established under a legally binding agreement.

Scope of Services — The activities and deliverables a service provider is required to perform.

Performance Standard — A defined level of service quality or operational achievement.

Key Performance Indicator (KPI) — A measurable indicator used to assess progress towards a specified objective.

Liability Clause — A contractual provision addressing responsibility for specified losses, damages, or obligations.

Indemnity — An undertaking by one party to compensate another for specified losses or liabilities under applicable conditions.

Force Majeure — A contractual concept addressing certain extraordinary events beyond the parties' control that may affect performance, subject to the agreement and applicable law.

Termination Clause — A contractual provision specifying circumstances and procedures for ending an agreement.

Contract Renewal — The extension or continuation of a contractual relationship under agreed conditions.

Dispute Resolution — The process of addressing contractual disagreements through agreed or legally available mechanisms.

Service Credit — A contractual adjustment or credit provided when specified service commitments are not met.

Contract Compliance — Adherence to the obligations and requirements established by an agreement.

Teaching content

KEY CONCEPT

Contract management and the purpose of contracts

The previous lesson, Logistics Strategy and Service-Provider Selection, ended with a decision: an external provider has been chosen to carry part of the logistics operation, on the strength of a specification, a scoring exercise and reference checks. The contract is where that choice becomes a working relationship, and contract management is what keeps the relationship working. Contract management is the administration and monitoring of contractual obligations throughout an agreement's life. A contractual obligation is a duty or responsibility established under a legally binding agreement: an undertaking to perform defined services, to pay agreed charges, to maintain insurance, to report performance, or to keep information confidential. On these definitions, contract management is not producing a signed document. It means everything that happens afterwards — seeing that both parties do what the agreement says, that performance is measured, that failures are corrected, and that the agreement itself is kept current as the business changes.

Contracts are needed in logistics service relationships because promises alone cannot carry an operation. A sound contract does several jobs at once.

1. It converts promises into defined duties, so both sides know precisely what is bought and sold.
2. It records what was agreed, because memories shorten and staff change while the contract endures.
3. It allocates responsibility and risk in advance: who insures, who answers for damage, who carries the cost of delay.
4. It sets the basis of payment — the rates and charges, and the service those prices assume.
5. It establishes accountability, giving the customer a structured route to demand correction instead of a daily quarrel.

Why, then, do logistics contracts fail? Far less often because of the law than because of the gap between what was promised and what was written. Three gaps recur. Vague scope: nobody

recorded exactly which activities were included, so work was assumed, omitted or performed twice. Unmeasurable commitments: the document says the provider will perform promptly and professionally, so no failure can ever be demonstrated. And no review mechanism: nothing in the agreement says when the parties will revisit volumes, targets and prices, so the contract hardens while the business keeps moving.

The professional view is therefore that a logistics contract is the relationship's operating system: not a filed certificate of trust, but the agreement the whole service runs on — the document that defines the service, measures it, prices it and provides for its repair.

KEY CONCEPT

Contractual obligations and performance standards are different things

Two ideas sit close together in every service contract and must be kept apart. A performance standard is a defined level of service quality or operational achievement — for example, that a stated share of deliveries arrives within the agreed window, or that stock records match physical counts within an agreed tolerance. A contractual obligation, by contrast, is a duty to do or not do something at all: perform the services in the scope, handle goods with care, maintain defined insurance, submit monthly reports, keep pricing confidential.

The difference shows in three ways.

1. Nature. An obligation is essentially binary: the insurance is maintained or it is not; the report was sent or it was not. A performance standard is graduated: service was better or worse than the agreed level, by a measured amount.
2. Evidence. Meeting or missing an obligation is established by documents and records. Meeting or missing a performance standard is established by measurement against a defined method and data source.
3. Consequences. A missed standard normally triggers the agreed service remedies — review, corrective action, service credits. An unmet obligation is a failure of contract compliance — adherence to the obligations and requirements established by an agreement — and is handled through escalation and, if it persists, the contract's dispute and termination provisions.

The two ideas also connect. Many contracts provide that persistent failure against a performance standard, left uncorrected, becomes a failure of obligation in its own right: the duty is not merely to deliver quickly but to manage the operation so that the agreed level is sustained. A manager reading a contract should therefore ask two separate questions of every clause. Is this a duty to do something, or a level to achieve? And what happens in this contract when it is not done, or not achieved?

KEY CONCEPT

The anatomy of a logistics services contract

Logistics service contracts differ in length and legal detail, but their working parts are broadly consistent. Taught structurally, the essential provisions are these.

1. Parties and definitions. The agreement names the contracting organisations and defines its recurring terms — the services, the premises, business days, data — so that later provisions can be short and precise.
2. Scope of services. What the provider will do, and just as importantly what it will not. Treated in full below.
3. Rates and charges. The prices for each service element, the assumptions behind them, and the rules for surcharges and indexation. Treated in full below.
4. Duration and renewal. How long the agreement runs and how it continues or ends. Treated in full below.
5. Liability and insurance. Who answers for loss or damage, to what agreed extent, and what insurance each party must hold. Treated in full below.
6. Confidentiality. Each party's undertaking to protect the other's commercial information — prices, customer lists, systems, volumes — exchanged in the course of the relationship.
7. Dispute resolution. The process of addressing contractual disagreements through agreed or legally available mechanisms. Well-managed contracts establish an escalation route first: operational discussion, then senior management, and only if those fail, the formal mechanisms. Most commercial disagreements should be settled on the operational ladder long before legal action is contemplated.
8. Termination and exit. The circumstances and procedures for ending the agreement, and the transition duties that follow. Treated in full below.

Around this frame, a service-level agreement usually rides as a schedule or annex: the contract provides the legal structure, and the SLA supplies the measurable heart. The provisions interlock. The scope defines what is measured; the measurements feed the remedies; persistent failure feeds the termination provisions; the renewal decision feeds everything again. A contract should therefore be read as one system, not as a stack of unrelated paragraphs — which is why contract managers insist on reviewing the whole document, not only the clause under dispute.

KEY CONCEPT

Scope of services: included work, exclusions, and assumptions

The scope of services is the set of activities and deliverables a service provider is required to perform. It is the clause every other clause leans on: the rates pay for the scope, the performance measures judge the scope, and the liability provisions allocate the risks of the scope. A vague scope therefore corrupts everything attached to it.

Good scope discipline, in the same spirit as the lesson on Preparing and Evaluating Logistics Quotations, has three parts.

1. Included activities, stated concretely. Not "warehouse services" but receiving, put-away, storage under defined conditions, picking, packing, dispatch, proof of delivery and returns handling — each with the locations, hours and volumes it assumes.
2. Exclusions, stated honestly. What the provider will not do is as important as what it will: pest control, cleaning, packing materials, after-hours work, haulage beyond named destinations.

Exclusions prevent scope creep — the slow accumulation of unpriced duties — and prevent the surprise of work nobody contracted for. An exclusion is not an insult; it is an honest boundary that pricing can respect.
3. Assumptions, stated openly. Contracts are priced against assumptions: weekly order volumes, average lines per order, operating hours, data formats. When the assumptions break, the pricing and the service break with them — which is why change management, treated later in this lesson, exists.

For international flows, one further link matters. The Incoterms 2020 rule chosen in each sales contract allocates the tasks, costs and risk-transfer points between buyer and seller — who arranges carriage, who insures, who clears the goods. The logistics services a provider is hired to perform must be read against those allocations, so that the service scope and the commercial terms describe the same division of work. Incoterms rules do this allocation only: they do not transfer ownership of the goods, fix prices, or themselves constitute a contract of carriage.

KEY CONCEPT

Rates, charges, and pricing provisions

The pricing provisions answer a deceptively simple question: what does each part of the service cost, and on what assumptions? The previous lesson noted that shippers typically buy capacity in two ways. A spot rate is a one-off price for a single movement at the current market level; a contract rate is an agreed price for a committed volume or period. Core lanes live on contract rates for predictability, and peaks are bought on the spot market. A logistics services contract brings the contract-rate side of that blend into formal terms.

Well-drafted pricing provisions share several features.

1. A rate structure with defined units. Storage per pallet per week, handling per order or per line, delivery per drop or per kilometre — with minimum charges and volume bands stated rather than implied.
2. Tariff references for time-based charges. The charges examined in the lesson on Storage, Detention, Demurrage, and Other Charges — storage beyond agreed free time, detention of vehicles while loading or queueing — arise from time, so the contract should reference the tariff or schedule that defines each charge and states clearly when the clock starts and stops. Surcharges on freight are handled with the same discipline, building on the concepts of the lesson on Shipping Charges, Freight Rates, and Surcharges.
3. Indexation rules, where used. Long agreements face moving fuel and market prices; a pre-agreed formula tied to a published measure adjusts charges without reopening the whole contract.
4. An invoice-checking basis. Charges are verified against the tariff and the measured service before payment — agreed rates matched to services actually delivered.

Unclear pricing provisions do not stay abstract: they surface as invoice disputes, withheld payments and quiet resentment. The test is the one this lesson applies throughout — could two strangers, reading only the document, compute the same charge for the same service?

KEY CONCEPT

Liability, insurance, and force majeure

When goods are lost or damaged, or a service failure costs the customer money, the question that decides the commercial outcome is: who carries this loss? Liability, insurance and related provisions answer that question in advance, calmly, instead of leaving it to a crisis.

A liability clause is a contractual provision addressing responsibility for specified losses, damages, or obligations. In logistics contracts such clauses commonly define how responsibility for loss of or damage to goods is approached, and may limit that responsibility to agreed extents or exclude defined categories of loss. An indemnity is an undertaking by one party to compensate another for specified losses or liabilities under applicable conditions — a duty to make good defined losses if defined events occur. The exact working of both depends on the law governing the agreement, which is precisely why such clauses are drafted by qualified advisers and managed by operations. The contract manager's job is not to interpret the law but to know what the clauses require and to see that conduct stays within them.

Insurance expectations connect to liability. Contracts typically require each party to hold defined classes of insurance appropriate to the risks it carries, to keep the cover in force, and to provide evidence of it. How cargo risk is allocated and insured is treated in the lesson on Cargo Insurance and Risk Allocation; the contract's job is structural — to state who must insure, against what, and who must prove it.

Force majeure is a contractual concept addressing certain extraordinary events beyond the parties' control that may affect performance, subject to the agreement and applicable law — events such as natural disasters or conflict, of the kind that has disrupted major trade routes in recent years. Such clauses typically provide that affected obligations may be suspended while the event lasts, with notice duties and, if the disruption is prolonged, a route to exit. These are the provisions nobody reads in a good year and everybody reads in a bad one.

KEY CONCEPT

Termination, renewal, and exit planning

Every contract has a life cycle, and the provisions governing its end deserve the same care as those governing its start.

A termination clause is a contractual provision specifying the circumstances and procedures for ending an agreement. Structurally, two routes recur. Termination for cause permits ending the agreement when the other party fails fundamentally — persistent unremedied service failure, breach of obligation, insolvency. Termination for convenience permits either party to end the agreement without fault, on agreed notice — a recognition that businesses change, and that a

relationship which no longer serves either side should end in order rather than in explosion. The clause also sets procedure: how notice is given, how long it runs, and what must happen during it.

Contract renewal is the extension or continuation of a contractual relationship under agreed conditions. Renewal can happen by drift — the agreement rolls on because nobody acted — or by decision, taken on evidence. The management discipline is to treat renewal as a fresh, evidence-based decision: performance against the SLA across the term, a market comparison from time to time, and the dependency risks the previous lesson described. Some agreements renew automatically unless notice is given by a stated date; a missed date can lock the parties in for another term, which is why renewal dates belong in the contract calendar rather than in anyone's memory.

Exit planning completes the set. A well-built contract provides for transition assistance when the agreement ends for any reason: handover of data and records, verified stock counts, reasonable support for a successor provider, and settlement of outstanding payments. These provisions are negotiated at the start precisely because no one wants to discuss them at the end — at signature there is goodwill, time and leverage; at exit there is usually none of the three. Exit terms also age: stock volumes grow, systems change, records multiply, so the exit plan is one of the items the review cycle, treated below, keeps current.

KEY CONCEPT

The service-level agreement: the measurable heart of the contract

A service-level agreement, or SLA, is an agreement defining specified service expectations, performance measures, and responsibilities. In logistics practice it usually takes the form of a schedule or annex to the main contract: the contract supplies the legal frame, and the SLA is its measurable heart — the part that turns "good service" into defined, counted expectations.

The previous lesson listed the SLA's components; this lesson treats them as a working structure.

1. Scope of services. What is being measured — the activities, locations and flows covered, consistent with the contract's scope clause.
2. Indicators and targets. The key performance indicators, each with a target: on-time in-full delivery, order accuracy, dock-to-stock time, inventory record accuracy, claim rates, response times.
3. Measurement and reporting arrangements. Who measures, from which system of record, by which method, and how often results are reported. This component decides whether the SLA can function at all.
4. Remedies and service credits. The pre-agreed consequences when targets are missed, and the escalation that follows persistent failure.
5. Governance and review cycle. The meetings that review performance, and the scheduled reviews that revisit targets, volumes and assumptions.

6. Exit and transition terms. What happens if the relationship ends — consistent with the contract's termination provisions.

Each component answers one question: what do we expect, how will we know, who measures, what happens when it is missed, and when do we revisit? Two failures destroy an SLA from opposite ends. An SLA without consequences is a wish list: targets with no remedies change nobody's behaviour. An SLA without a defined data source is an argument: if the parties measure differently, every month's score is a negotiation rather than a fact. The measure, the source and the consequence stand or fall together — the drafting discipline the next key concept examines element by element.

KEY CONCEPT

Writing measurable commitments: indicators, targets, and measurement sources

A key performance indicator is a measurable indicator used to assess progress towards a specified objective. In an SLA, the indicators are usually drawn from the service measures of the lesson on Logistics Performance Indicators and KPIs: on-time in-full delivery — computed as orders delivered both on time and in full, divided by total deliveries, multiplied by 100, and strict enough that an order fails if it is late or short; order accuracy; dock-to-stock time, the elapsed time from arrival at the receiving dock until goods are available to pick; inventory record accuracy; claims and damage ratios; and the timeliness of documents such as proof of delivery.

The drafting discipline is this: write every commitment so that two strangers, given the same data, would score it identically. A commitment that depends on whose memory is consulted is not a commitment. Each measured commitment therefore needs five elements.

1. The indicator and its formula, defined so that counting is mechanical.
2. The counting window — monthly, per lane, per site — with any exclusions stated and rule-bound, for example delays documented as customer-caused under an agreed procedure.
3. The target, set from evidence. Targets are tied to the parties' own measured baseline — set just beyond it, in the direction the service promise requires — not borrowed from another company's benchmark, which no one can verify applies here.
4. The data source and the measurer. Which system of record supplies the numbers — the provider's warehouse or transport system as described in the lesson on Transport and Warehouse Management Systems, or the shared tracking described in the lesson on Shipment Visibility, Tracking, and Automation — and who computes the score.
5. The reporting cadence — when results are published, and to whom.

Two cautions complete the discipline. Few measures, well chosen, outperform a long list nobody reads; and because people manage what is measured, a badly designed measure gets gamed — reward only on-time arrivals and goods will be held back to make the window. The measure must reflect the promise the customer actually bought.

KEY CONCEPT

Remedies and service credits: consequences for missed performance

A service credit is a contractual adjustment or credit provided when specified service commitments are not met — in effect, a pre-agreed price reduction for a period in which the service fell short of the promise. Its purpose is not punishment. Pre-agreed consequences price the failure in advance, keep both sides out of improvised penalties, and give the provider a financial reason to invest in correction rather than explanation.

Mature SLAs arrange consequences as an escalation ladder, stepped to the persistence and seriousness of failure. The steps, in the order they are normally written:

1. Review. The miss is visible — reported in the agreed format — and discussed at the operational meeting. Many misses end here: explained, logged, corrected.
2. Corrective-action plan. Where failure repeats, the provider documents the root cause, the fix, the owner and the dates, and progress against the plan is tracked at every meeting until closure.
3. Service credits. Where misses persist beyond the agreed threshold — commonly, consecutive months below target — credits are applied per the schedule in the SLA. The schedule is stated in advance: which indicator, how far below target, what adjustment.
4. Exit trigger. Persistent failure that survives correction becomes grounds for termination for cause under the contract — the ladder's last rung, connected to the termination provisions treated earlier.

The ladder's value lies in being followed as written. Improvised responses — a shouted threat one month, a waived credit the next — teach both sides that the document is decorative. Applied as written, the ladder converts failure from conflict into procedure: the customer enforces without escalating personally, and the provider corrects without renegotiating its survival each month. Some agreements add the mirror image — incentives for sustained outperformance — and that is optional. The remedies ladder is not. An SLA whose consequences have never been applied should prompt a question: is the service excellent, or is nobody measuring?

KEY CONCEPT

Governance and monitoring: keeping the contract alive

Contract compliance is adherence to the obligations and requirements established by an agreement. Compliance is demonstrated, not assumed: performance is reviewed against the SLA with evidence, using the measured data of the lesson on Logistics Performance Indicators and KPIs rather than impressions exchanged in corridors.

The working rhythm has two clocks.

1. The operational clock. Regular meetings — commonly weekly or monthly — where the period's results are tabled, misses are explained, the issue log is worked through, and corrective actions are tracked to closure. An issue log is deliberately simple: every raised problem is

recorded, owned, dated and closed. The pattern that matters is closure — issues that are logged and never closed are the leading indicator of a relationship that will end badly.
2. The review clock. Periodic business reviews — commonly quarterly — at a more senior level: trends rather than incidents, root causes rather than symptoms, and the health of the relationship itself — candour in meetings, responsiveness when things go wrong, stability of the provider's staff. This is the rhythm the previous lesson introduced as provider governance; under a contract it becomes the mechanism by which the SLA actually operates.

Alongside the rhythm stand two practices. Change management: volumes shift, lanes shift, customers arrive and leave — so a variation process allows the parties to amend scope, volumes, rates or targets by documented, signed agreement, keeping the contract true without tearing it up. Undocumented arrangements — the informal favour, the side email — corrode exactly the clarity the contract exists to create. And the scheduled review of the whole agreement — typically annual — at which assumptions, targets, rates and exit plans are revisited against evidence. Reviews are maintenance, not ceremony: the review cycle is the contract's heartbeat, and a contract whose reviews are skipped goes stale silently. This discipline of structured, evidence-based review is also where contract management meets the continuous improvement treated in this module's final lesson, Sustainability, Ethics, and Continuous Improvement.

WORKED EXAMPLE

Rewriting a vague commitment: a consumer-goods distributor in Kumasi

A distributor of packaged consumer goods based in Kumasi contracts a regional provider to run its warehouse and deliver to wholesale customers in Kumasi, Sunyani and Tamale. The provider's draft contains this commitment: "The provider will deliver orders on time at competitive rates and manage the warehouse to a high standard." The distributor's contract manager rewrites it before signature. All figures in this example are illustrative for teaching purposes.

Step 1: Read the clause as a stranger would. When is "on time"? Competitive against what? High by which standard? Every phrase requires a judgement nobody has defined, so no failure could ever be demonstrated — the gap between promised and written, captured in one sentence.

Step 2: Decompose the promise into service elements. Delivery reliability; order correctness; speed of putting received goods away; trustworthiness of stock records. Four elements, four commitments.

Step 3: Give each element an indicator and formula. On-time in-full delivery: orders delivered both on time and in full, divided by total deliveries, multiplied by 100 — strict, so an order that is early but short still fails. Order accuracy: orders delivered without item or quantity error, divided by total orders, multiplied by 100. Dock-to-stock time: hours from vehicle arrival at the receiving dock until goods are available to pick. Inventory record accuracy: counted locations matching the record within tolerance, divided by locations counted, multiplied by 100.

Step 4: Set targets from the baseline. A three-month transition period is measured first. Illustrative results are on-time in-full of 88 per cent, order accuracy of 97 per cent, dock-to-stock of 30 hours and record accuracy of 96 per cent. Targets are set just beyond the baseline — illustratively, 92 per cent, 98 per cent, 24 hours and 98 per cent for the first contract year — not copied from an industry table nobody can verify.

Step 5: Name the data source, the measurer and the cadence. Dispatch and delivery confirmations come from the provider's warehouse system; proof-of-delivery timestamps fix delivery times; the provider measures and reports monthly to the distributor, and the operational meeting reviews the results.

Step 6: Attach the remedy. If monthly on-time in-full falls below target for two consecutive months, the SLA's service-credit schedule applies and a corrective-action plan is required — the ladder treated in the key concept on remedies and service credits.

What does this example demonstrate?
The before-and-after of drafting discipline. Key lesson: the difference between a promise and a commitment is a number, a method, a source and a consequence.

WORKED EXAMPLE

The escalation ladder applied: an electronics distributor in Singapore

A consumer-electronics distributor in Singapore contracts a regional provider for warehousing and next-day island-wide delivery. The SLA sets on-time in-full delivery at a target of 95 per cent, measured monthly from the provider's system records and proof-of-delivery data, with a service-credit schedule and an escalation ladder agreed in advance. All figures in this example are illustrative for teaching purposes.

Step 1: Month one — a miss. On-time in-full is reported at 92 per cent. The miss appears at the monthly operational meeting, as the ladder's first step requires: the provider explains a driver shortage, the miss is logged, and corrective action is agreed for the rosters. No credit yet — one month below target is review territory under the agreed ladder.

Step 2: Month two — the pattern. On-time in-full is 93 per cent. Two consecutive months below target is the threshold written in the SLA, so the ladder is followed as written. The provider submits a corrective-action plan within the agreed period: root cause identified as picking backlogs after promotional volume spikes, the fix is extra picking capacity in promotion weeks, and the owner and dates are named. The service credit for the month is applied to the next invoice per the schedule — without argument, because the schedule was agreed when nobody was failing.

Step 3: Month three — the trend. On-time in-full is 94 per cent: still below target, so the credit applies again under the schedule, but the weekly reviews show the plan working, with promotion-week misses falling. The ladder distinguishes persistence from recovery: the exit trigger requires sustained failure with no effective plan, not a recovery in progress.

Step 4: Month four — recovery verified. On-time in-full reaches 96 per cent. The issue is closed at the operational meeting; the next quarterly business review records the episode, the amended picking process and the updated issue log. Nothing was renegotiated mid-failure; the document did the work.

Step 5: The road not taken. Consider the same two months managed without a ladder: an angry telephone call in month one, all payment withheld in month two, a threat of termination in month three — while the picking backlog quietly continued. Improvisation converts a fixable backlog into a relationship crisis.

What does this example demonstrate?
Consequences agreed in advance let both parties walk a failure back to health. Key lesson: an SLA's value is proven not in the month the targets are met but in the month they are missed — followed as written, the ladder turns failure into procedure instead of conflict.

WORKED EXAMPLE

The annual review that mattered: a frozen-fish processor in Tema

A frozen-fish processor in Tema holds a three-year contract with a cold-chain provider: store frozen fish at defined temperatures in the provider's cold store, and distribute to retail depots in Accra, Kumasi and Takoradi. The SLA's targets and the rate schedule were built on agreed baseline volumes — illustratively, 400 pallets stored and three dispatch runs a week. All figures in this example are illustrative for teaching purposes.

Step 1: Growth arrives. The processor wins two supermarket chains, and volumes rise by roughly 40 per cent within a year, with some weeks running double dispatches. Nobody reneges on anything; the contract simply no longer describes the operation.

Step 2: The assumptions break quietly. The cold store fills towards capacity, so receiving slows and dock-to-stock time stretches beyond its target. Dispatch density rises above the volumes the rate bands assumed. On-time in-full begins to slip as the provider juggles space. Both parties are quietly losing: the provider absorbs costs the rates never priced, while the customer's measured service decays.

Step 3: The scheduled annual review happens. It would have been easy to skip — the relationship is friendly and both sides are busy. Instead, the review goes ahead with evidence tabled: twelve months of volumes, capacity use and SLA results.

Step 4: The contract is re-scoped through its variation process. The parties sign a documented amendment. Volume commitments and guaranteed cold-store capacity are restated for the new baseline. Additional dispatch capacity is added. The dock-to-stock target is reset to the new reality and then stepped down as process changes land. Rate bands are re-cut for the higher volumes — the processor pays a lower rate per pallet at the new scale while the provider commits the capacity. The exit provisions are refreshed too, because stock holdings and data records have grown with the business.

Step 5: The counterfactual. Had the review been skipped, each side would have kept silently repricing the relationship — the provider in thinner service, the customer in quieter resentment — until the failure surfaced as a crisis, with exit negotiated in anger at the worst possible moment.

What does this example demonstrate?
Review as maintenance, not ceremony. Key lesson: the review cycle is the contract's heartbeat — a contract kept current costs a meeting; a contract gone stale costs the relationship.

IMPORTANT NOTE

An SLA without consequences is a wish list; without a data source it is an argument

Two defects kill service-level agreements from opposite ends, and both are common. The first is the SLA without consequences: a page of targets — delivery on time, accuracy high, response swift — with no remedy attached to any of them. It reads well and changes nothing, because nothing happens when a target is missed; it is a wish list, not an agreement. The second is the SLA without a defined data source: targets exist and remedies exist, but nobody specified which system supplies the numbers or who measures. The first missed month becomes a dispute about arithmetic instead of a fact — one side's system says 93 per cent, the other's says 96, and the difference is the argument.

The remedy is structural, and it is the one this lesson has built: every commitment needs its indicator, its formula, its source, its measurer and its consequence, or it does not belong in the agreement. A related warning completes the set: the signed-but-shelved contract. Some organisations manage their contracts with a single act — the signature — and a file copy. Measuring is not managing: the reports must be read, the meetings must happen, the ladder must be followed, or the machinery exists only on paper. The test of contract management is not the quality of the document but the routine that operates it.

IMPORTANT NOTE

A missed target is not the same as a broken obligation

This lesson's most practical distinction bears consolidating, because practitioners blur it weekly. A missed performance target is a measured service failure, answered through the SLA's machinery: review, corrective action, service credit. A broken contractual obligation — insurance allowed to lapse, reports never sent, confidential prices disclosed — is a failure of contract compliance, answered through escalation and, if it persists, the dispute and termination provisions. The two can move in opposite directions: a provider can hit every monthly indicator while quietly breaching obligations, which is why contract compliance reviews cover duties as well as scores; and a missed target handled properly through the ladder is not automatically a legal rupture — the ladder exists precisely so that ordinary failures stay management matters rather than becoming disputes.

One boundary must also be drawn honestly. This lesson teaches contract management as a commercial and operational discipline: what the provisions are for, how commitments are made measurable, how performance is governed. It teaches no country's law, and its structural descriptions are not legal advice. Contracts are drafted and interpreted under applicable law by qualified advisers; the logistics professional's contribution is to know what the agreement requires operationally, to keep evidence, and to recognise when a question has left the operations meeting for the lawyers.

IMPORTANT NOTE

Exit clauses are written at the start because no one will discuss them at the end

Exit and termination provisions look morbid at signature, which is exactly why they must be agreed then. At the start of a relationship there is goodwill, time and negotiating balance; at the end there are usually none of the three. A transition worked out calmly at signing — data and records handed over, stock counted and released, reasonable assistance given to a successor — becomes an improvised standoff when first discussed during a failure. The same logic keeps exit terms under maintenance: stock holdings grow, systems change, records multiply, so the exit plan agreed three years ago may no longer describe what would actually have to be handed over.

The wider habit these provisions teach is that contracts are living documents. The review cycle keeps assumptions current; the variation process keeps scope and rates true; the exit plan stays usable. A contract allowed to go stale does not merely underperform — it becomes a continuity risk, because a business depending on an outdated agreement has no tested answer for the day the agreement and the reality finally part company. That continuity question — how a logistics operation keeps running through disruption, including the disruption of a failing provider relationship — is the subject of the next lesson, Business Continuity and Supply Chain Resilience.

Practice questions (Lesson Checks)

Practice Question 1

A manager of a Kumasi-based distribution company receives a draft logistics contract containing the commitment: "The provider will deliver orders promptly and keep storage charges competitive." She asks her team how the commitment should be strengthened before signature.

Which option best describes the change that would make this commitment manageable?

1. The commitment should be shortened, because shorter clauses leave less room for disagreement about their meaning.

2. The commitment should state the indicator, the measurement formula, the system the data comes from, and the remedy that applies when the target is missed. ✓

3. The commitment should be reviewed by a lawyer, who will define "promptly" and "competitive" in legally precise language.

4. The commitment should be moved into a cover letter, so the main contract stays simple and easy to read.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

A commitment becomes manageable when a stranger, given the same data, would score it identically every time. That requires the full drafting discipline of this lesson: a named indicator with its formula — for example, on-time in-full delivery computed from delivery confirmations — a defined data source and measurer, a target set from the parties' own baseline, and a pre-agreed remedy when the target is missed. A promise carrying a number, a method, a source and a consequence can be tracked, enforced and improved.

Answer 1 fails because brevity is the problem, not the cure: a short vague clause is still unmeasurable. Answer 3 mistakes the discipline: legal review serves legal effect, but no amount of legal polish makes "promptly" countable — what is missing is measurement, not wording. Answer 4 weakens the commitment further by moving it outside the enforceable agreement.

The takeaway: if a commitment cannot be measured from a defined source, it is a preference, not a contract term.

Practice Question 2

A frozen-food importer in Tema holds a warehousing and distribution contract whose service-level agreement sets an on-time in-full target, a service-credit schedule, and an escalation ladder: review, corrective-action plan, credits for consecutive misses, and an exit trigger reserved for persistent failure with no effective corrective plan. The provider has now fallen below the on-time in-full target for two consecutive months, and a corrective-action plan is in place with evidence of recovery under way.

Which statement best describes how the customer should respond?

1. Terminate the contract immediately for cause, because two missed months prove the provider cannot recover.

2. Apply the escalation ladder as written: apply the service credit for the consecutive misses, track the corrective-action plan to closure, and reserve termination for persistent failure with no effective plan. ✓

3. Waive the credit this month to preserve goodwill, and ask the provider to try harder next quarter.

4. Renegotiate the on-time in-full target downwards so the provider can meet it while the recovery continues.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The ladder's value lies in being followed as written. Two consecutive months below target is exactly the threshold the SLA anticipated, so the agreed consequences follow: the service credit is applied per the schedule, the corrective-action plan is tracked at the meetings until it closes, and recovery is verified with the measured data. The exit trigger stays reserved for its defined condition — sustained failure that survives correction — which this scenario, with a plan in place and recovery evident, does not meet.

Answer 1 bypasses the agreed machinery and contradicts the contract's own definition of cause. Answer 3 improvises: waiving credits teaches both sides that the document is decorative, and goodwill is preserved by procedure, not by exceptions. Answer 4 quietly rewrites the deal in the provider's favour, destroying the discipline that targets be set from evidence.

The takeaway: consequences agreed in advance only work if they are applied as agreed — in failure as reliably as in success.

Practice Question 3

A pharmaceutical distributor in Accra reviews its cold-chain distribution contract at year end. The provider's on-time in-full and temperature-excursion indicators have met their targets every month. However, during the year the provider stopped submitting the agreed monthly temperature-report summaries, and allowed a required insurance certificate to lapse for part of the year.

Which statement best describes this situation?

1. The contract is in full compliance, because performance against the indicators is the only measure of compliance in a service contract.

2. The reporting and insurance duties are courtesies outside the contract, so only the missed indicators could ever carry consequences.

3. The performance targets were met, but contractual obligations were not: compliance covers duties as well as scores, so the reporting and insurance failures require escalation even though the indicators are green. ✓

4. The situation shows the SLA is worthless and should be replaced with a purely legal agreement enforced through the courts.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

This lesson's central distinction is between contractual obligations — duties to do or not do something, such as submitting agreed reports and maintaining required insurance — and performance standards, the measured levels the SLA tracks. The provider met the standards but failed the duties. Contract compliance is adherence to all the obligations and requirements of the agreement, so a green scorecard does not certify a compliant contract: obligation failures are established by records and escalation, not by the indicator report.

Answer 1 fails because it collapses compliance into measurement, ignoring exactly the duties that would protect the distributor if a cold-chain claim ever arose. Answer 2 is wrong because reporting and insurance duties sit inside the agreement as obligations, with their own consequences. Answer 4 overreacts: the SLA and the obligations machinery work together, and the response to obligation failure is escalation under the contract, not abolition of measurement.

The takeaway: judge a contract relationship on duties kept as well as targets hit.

Fact-check appendix (audit notes — not for entry)

Definitions of contract management, service-level agreement, contractual obligation, scope of services, performance standard, key performance indicator, liability clause, indemnity, force majeure, termination clause, contract renewal, dispute resolution, service credit, and contract compliance follow this lesson's key-term definitions — source: lesson brief 15.2, curriculum key terms.

The SLA component structure (scope of services; indicators and targets; measurement and reporting arrangements; remedies and service credits; governance and review cycle; exit and transition terms) — source: curriculum research sheet logistics-scm-frameworks.md, section 10 (industry practice; taught as components, not a fixed standard).

Spot rate as a one-off price at current market and contract rate as an agreed price for committed volume or period; shippers typically blend both — source: research sheet logistics-scm-frameworks.md, section 10.

OTIF formula (orders delivered both on time and in full, divided by total deliveries, multiplied by 100) and its strictness (an order fails if late or short) — source: research sheet logistics-scm-frameworks.md, section 7 (MetricHQ).

Dock-to-stock time and inventory record accuracy as standard logistics service indicators — source: research sheet logistics-scm-frameworks.md, sections 7 and 4.

Incoterms 2020 rules allocate tasks, costs and risk-transfer points between buyer and seller, and do not transfer ownership, fix prices, or constitute a contract of carriage — source: research sheet logistics-scm-frameworks.md, section 10.

The escalation ladder (review, corrective-action plan, service credits, exit trigger) is taught as a general commercial structure of escalating remedies, consistent with the remedies/service-credits component in research sheet section 10; it reflects general business practice, not any jurisdiction's law.

Setting targets just beyond the parties' own measured baseline rather than borrowed benchmarks — consistent with the lesson 13.4 treatment of Logistics Performance Indicators and KPIs (curriculum drafts, module 13) and general business practice.

All numeric targets, percentages, volumes and time figures in the worked examples are illustrative for teaching purposes (specification rule 5.1): no amounts, statutory deadlines or free-time periods are stated as fact.

Lesson 15.3 — Business Continuity and Supply Chain Resilience

Summary

This lesson examines how organisations prepare for, respond to, and recover from disruptions affecting logistics and supply chain operations. Learners will explore business continuity planning, supply chain risk assessment, contingency arrangements, alternative suppliers, transportation disruptions, and recovery strategies. The lesson emphasises the importance of identifying critical activities and developing practical plans to maintain essential services during unexpected events.

Learning objectives

- Define business continuity and supply chain resilience.
- Identify common disruptions affecting logistics and supply chain operations.
- Explain the importance of identifying critical logistics activities.
- Describe the purpose of business continuity plans.
- Distinguish between risk prevention, contingency planning, and disaster recovery.
- Identify strategies for reducing dependence on vulnerable suppliers or transportation routes.
- Explain how alternative transport arrangements and inventory buffers support resilience.
- Describe the importance of communication during operational disruptions.
- Identify basic measures used to evaluate business continuity readiness.
- Develop a simple contingency plan for a major logistics disruption.

Key terms

Business Continuity — The capability of an organisation to maintain or resume essential activities during and after a disruption.

Business Continuity Plan (BCP) — A documented plan describing how an organisation will respond to disruptions and maintain or restore critical operations.

Supply Chain Resilience — The ability of a supply chain to prepare for, respond to, adapt to, and recover from disruptions.

Operational Disruption — An event that interrupts or significantly affects normal business activities.

Risk Assessment — The process of identifying risks and evaluating their likelihood and potential consequences.

Business Impact Analysis (BIA) — The assessment of how disruptions to business activities may affect organisational operations and objectives.

Contingency Planning — The preparation of alternative actions for responding to unexpected events.

Disaster Recovery — The restoration of systems, facilities, or operations following a major disruptive event.

Recovery Time Objective (RTO) — The targeted maximum period within which a disrupted process or system should be restored.

Recovery Point Objective (RPO) — The maximum acceptable amount of data loss measured in time before a disruption.

Supplier Diversification — The use of multiple suppliers to reduce dependence on a single source.

Redundancy — The availability of additional resources, systems, or capacity that can support operations when primary resources fail.

Single Point of Failure — A component or dependency whose failure can cause a wider system or operation to stop functioning.

Resilience Testing — The evaluation of an organisation's ability to respond to and recover from simulated or actual disruptions.

Teaching content

KEY CONCEPT

Disruption is the norm: business continuity and supply chain resilience

The lesson on Managing Shipment Delays, Damage, and Discrepancies dealt with incidents one at a time: a delayed consignment, a damaged pallet, a shortage against a delivery note. This lesson rises a level: not how to handle the incident in front of us, but how to prepare an entire logistics operation for disruption before it arrives — decided calmly in advance rather than improvised under pressure.

An operational disruption is an event that interrupts or significantly affects normal business activities. The record of recent years teaches that such events are a normal condition of international logistics, not an aberration. A pandemic slowed ports and freight lanes for roughly two years. A single grounded vessel closed the world's best-known shipping canal for the better part of a week. Attacks in one sea area pushed the largest shipping lines onto a route around an entire continent. Events that begin far away do not stay far away: they propagate through

freight rates, vessel schedules and lead times until they arrive in every warehouse and at every border post.

Business continuity is the capability of an organisation to maintain or resume essential activities during and after a disruption. Supply chain resilience is the wider quality: the ability of a supply chain to prepare for, respond to, adapt to, and recover from disruptions. The two ideas differ in scope. Continuity is what one organisation can do; resilience is what the whole chain — suppliers, carriers, terminals, warehouses, customers together — can do.

Notice the verbs. Prepare comes before respond, and respond before recover. That ordering is the lesson's argument: the strategic response to disruption is designed before the event. A business that first meets its disruption on the day it happens is not responding; it is discovering.

KEY CONCEPT

The disruption landscape: categories of risk and the recent record

Preparation begins with knowing what can go wrong. The first stage of a risk assessment — the process of identifying risks and evaluating their likelihood and potential consequences — is simply listing them, and a set of six categories keeps the list honest.

1. Supply-side risks
A key supplier fails — fire, insolvency, raw-material shortage, quality stoppage — and the flow halts before it ever reaches a truck.

2. Transport risks
Ports congest, vessels delay, corridors close, carriers fail, fuel spikes. Most of what can delay a consignment lives here.

3. Customs and regulatory risks
New requirements, system outages, staff shortages or tightened inspections hold goods at the border after they have travelled halfway round the world.

4. Financial risks
Currency swings, customer failures and sudden cost increases can make an otherwise sound flow unaffordable.

5. Security risks
Theft, pilferage, deliberate damage and attacks on cargo or infrastructure — including attacks on shipping itself.

6. Natural risks
Floods, storms, drought and disease disrupt facilities, roads and harvests; some, such as an annual rainy season, are predictable enough to plan around.

For each risk, likelihood and impact are judged — usually qualitatively, high, medium or low — and the pairing says where preparation is worth its cost.

The recent record shows how these categories combine. From early 2020 through roughly 2021 and 2022, the COVID-19 pandemic brought port congestion to the global trading system: demand shifted between products and regions, labour disruptions slowed terminals, vessels and containers piled up in the wrong places, and freight rates on major lanes reached record levels. In March 2021 the Ever Given ran aground in the Suez Canal on the 23rd and was freed on the 29th — six days and seven hours during which one of the world's principal trade arteries was closed, with at least 369 vessels queued by 28 March and over 400 by reopening. And from late 2023, attacks on shipping in the Red Sea and Gulf of Aden led major container lines, from December 2023, to reroute widely around the Cape of Good Hope, substantially lengthening Asia–Europe voyages.

None of these events consulted any individual business first — which is exactly why the response must be designed in advance.

KEY CONCEPT

Critical activities first: the business impact analysis

A continuity effort that tries to protect everything protects nothing well. The discipline that brings order is the business impact analysis — the assessment of how disruptions to business activities may affect organisational operations and objectives. Its question is blunt: if this activity stopped today, what would happen, how soon would it hurt, and how much would it cost?

Applied to a logistics operation, the analysis sorts activities into three groups.

1. Activities that must continue without interruption
Order-taking, so customers can still place orders and be told the truth; the critical stock lines whose absence halts customers' own work — medicines for a pharmacy's customers, cement for a construction site; and delivery to key customers whose businesses depend on the flow. These are the activities a disruption may not stop for long.

2. Activities that can slow down
Non-critical replenishment, routine stock movements, internal reporting. They can run late for days while the essentials hold.

3. Activities that can stop
Projects, convenience services, discretionary spending. Recognising these honestly is what funds the protection of the first group.

Each critical activity then has its dependencies mapped — the people, systems, suppliers, transport links and documents it cannot run without — because it is exactly as resilient as its weakest dependency. The analysis ends by setting recovery priorities: an explicit ordering of what recovers first, second and third, agreed by management before any event, so that a disruption does not open with an argument about what matters.

The importance of identifying critical activities is hard to overstate: it converts a vague wish to keep the business running into a ranked, fundable list of promises, and every later element of this lesson hangs off that list.

KEY CONCEPT

The concentration map: finding single points of failure

A single point of failure is a component or dependency whose failure can cause a wider system or operation to stop functioning. Concentration is how modern supply chains earn their efficiency: one supplier at scale, one corridor with consolidated volumes, one terminal under a single handling agreement, one warehouse holding all the stock, one system holding all the records. Each arrangement saves money every day it works. The map exists because the same arrangements can stop the operation in a single afternoon when they fail.

The concentration map is drawn by tracing, for each critical activity, every dependency it rests on — and circling the ones with no alternative. The questions are mechanical, which is their virtue. Which suppliers can supply this line? If the answer is one, circle it. Which corridors and ports does the inbound flow actually use? If every container rides one route into one terminal, circle both. Which warehouse holds the safety stock? Which system holds the stock records, and where does its data live? Which person holds the only working knowledge of the clearance process? People, as much as ports, are single points of failure.

The lesson on Logistics Strategy and Service-Provider Selection examined dependence on a single service provider and the diversification that answers it. The concentration map widens that lens from providers to the whole chain: a business can hold contracts with three carriers and still be fatally concentrated if all three sail the same corridor into the same terminal.

Concentration is not automatically a mistake — sometimes no alternative exists, and redundancy costs money, as later sections weigh. The mistake is never drawing the map. A dependency nobody has named is a decision nobody has taken; the audit of single points of failure is simple, cheap and renewed annually.

KEY CONCEPT

The business continuity plan: purpose and contents

A business continuity plan, or BCP, is a documented plan describing how an organisation will respond to disruptions and maintain or restore critical operations. Its purpose is to move the expensive thinking — what to protect, in what order, by what means, under whose authority — from the middle of a crisis to a calm day months before it.

Effective continuity plans for logistics operations carry the same backbone.

1. Activation

Who declares that a disruption exists, on what evidence, and with what authority to spend and decide. Without this, the first hours are lost establishing permission.

2. The critical list and recovery priorities
The business impact analysis in writing: which activities must continue, which may slow, and the agreed order of recovery.

3. Fallback arrangements
For each critical activity, the documented alternative: the second supplier with contacts and terms, the alternative corridor, the manual procedure that replaces the system, the second warehouse.

4. Roles and contacts
Who leads, who re-routes, who talks to customers, who talks to suppliers and authorities — and how each is reached at midnight, current telephone numbers being among the most decayed information in any organisation.

5. Communication material
Prepared message templates for customers, suppliers and staff, because words chosen mid-crisis are the words most likely to be wrong.

A plan unwritten is a wish. But a written plan in a drawer is barely better, because continuity is performed by people who must know their part in advance. The plan must therefore be communicated: everyone named in it reads and accepts their role, key customers and suppliers know what to expect if disruption comes, and the commitments line up with the contracts of the previous lesson on Contract Management and Service-Level Agreements. It also depends on records — the offline stock position, the dispatch papers that let a manual release proceed — built through the ordinary discipline of warehouse documentation and stock reconciliation.

KEY CONCEPT

Prevention, contingency, and recovery: three layers of response

Three disciplines are easily blurred and must be kept apart, because they answer different questions on different sides of the event.

1. Risk prevention
Prevention acts before the event to make it less likely or less damaging: clearing warehouse drainage before the rains, maintaining vehicles so they do not fail in transit, training staff, controlling access against theft. Insurance belongs in this layer as risk transfer — the lesson on Cargo Insurance and Risk Allocation showed how a policy carries the financial consequence of events the business cannot prevent. But note what insurance does and does not do: it compensates loss afterwards; it does not keep goods moving, customers served or systems running.

2. Contingency planning
Contingency planning is the preparation of alternative actions for responding to unexpected events. It also happens before the event, but its product is not a reduced likelihood — it is a ready alternative: the qualified second supplier, the documented alternative route, the manual

workaround. Contingency planning assumes the event will happen anyway, and answers in advance the question, then what?

3. Disaster recovery
Disaster recovery is the restoration of systems, facilities, or operations following a major disruptive event. It is the after-discipline: rebuilding the damaged warehouse, restoring the collapsed system, standing operations up again. Its two great measures are the recovery time objective — the targeted maximum period within which a disrupted process or system should be restored — and the recovery point objective, the maximum acceptable amount of data loss measured in time before a disruption: not only how fast the system returns, but how much of its information survives.

The layers interlock. Prevention shrinks the odds; contingency shortens the fall; recovery sets the pace of getting up. A business strong in only one layer is prepared for only one half of a disruption. The remainder of this lesson lives mostly in the middle layer — the strategies that give a contingency plan its alternatives.

KEY CONCEPT

Supplier diversification and redundancy: the insurance logic

Supplier diversification is the use of multiple suppliers to reduce dependence on a single source — the direct answer to a circled name on the concentration map. Redundancy is the wider principle it applies: the availability of additional resources, systems, or capacity that can support operations when primary resources fail. A second supplier, a second route, a standby generator and a cross-trained colleague are the same idea in different clothes: a way already held open, at a cost, for the day the primary way closes.

Dual sourcing is usually the first strategy considered for critical inputs, and it should be priced honestly. The second supplier almost always costs more per unit — it lacks the primary's scale or geography — so the operation pays a standing premium for capacity it does not use in normal times. That premium should be understood exactly as an insurance premium is understood in the lesson on Cargo Insurance and Risk Allocation: a modest, certain, regular payment bought against a rare, severe, uncertain loss. Nobody calls insurance wasteful because the fire never comes.

Two disciplines make diversification real rather than nominal.

1. Qualify the second supplier properly
A name on a list is not a second source. The alternative must pass the same evaluation discipline as the first — capability, quality, capacity, reliability — and ideally carry some real volume, even a modest committed share, so that the relationship, tooling and performance knowledge stay warm. A supplier whose first order in three years arrives mid-crisis will not save anyone.

2. Diversify deliberately, not equally
Dual sourcing does not require splitting volumes half and half. The sensible pattern is a primary

share for cost, plus a protected minority share that keeps the alternative alive. The trade-off — higher unit cost, two relationships to manage, smaller consignments — is the premium, weighed numerically in this lesson's final worked example.

KEY CONCEPT

The inventory buffer: resilience carried on the shelf

The second resilience strategy is buffer stock: inventory held not because demand is expected, but because supply may fail. The lesson on Inventory Optimisation and Service Levels distinguished cycle stock, which works down and is replenished, from safety stock, the buffer against demand and lead-time variability. Resilience adds a third job: the strategic buffer against supply disruption — extra depth on the critical lines, sized not by demand statistics but by how long the chain behind them could take to recover.

The logic connects directly to that earlier lesson's service-level thinking. A buffer exists to let the business keep serving customers through a gap, so its depth depends on how long the gap could last: a week's interruption from a delayed vessel needs less cover than a month's interruption from a failed supplier. The shorter and more reliable the replenishment chain, the smaller the buffer required — which is why buffer stock and supplier diversification are complements. A second supplier shortens the worst-case gap, and so shrinks the stock needed to survive it.

The trade-off is the one that lesson made familiar: every carton of buffer carries holding cost. Capital sits tied up, storage space is occupied, and the goods themselves risk deterioration, obsolescence, expiry or theft. A buffer is not a free good, and an operation that buffers everything will find its working capital and its warehouse both full. The discipline is segmentation: buffer the critical lines identified in the business impact analysis — the medicines, the fasteners, the spare part that stops the line — and hold the commodity lines lean, accepting their exposure because their impact is small and their replacement is easy.

KEY CONCEPT

Visibility: early warning that buys time

The third strategy is visibility, and its value in continuity is measured in time. End-to-end visibility — knowing where orders, stock and shipments stand across the whole chain — converts a disruption from a surprise into a decision. The lesson on Shipment Visibility, Tracking, and Automation built this capability: milestone events, condition sensors and automated exception alerts that surface the one shipment among thousands that has stopped moving. Event-driven visibility is the early-warning system of supply chain resilience.

Its contribution is clearest in the arithmetic of response. A port congestion alert received on the day a vessel sails leaves weeks of options: re-book, re-route, re-prioritise the critical lines, warn customers honestly. The same news received on the day the vessel fails to arrive leaves almost none. Early warning does not prevent the disruption; it purchases the time in which the

contingency plan can actually be used — and many plans fail not because their alternatives were wrong, but because they were activated too late.

Visibility works at two depths. Operational visibility watches the flows: is the vessel on schedule, has the container been discharged, has the truck left. Structural visibility watches the shape of the chain itself: which suppliers remain open, which corridors still function, where the buffer stands today against the critical list. The concentration map of the earlier section is precisely this second kind of visibility, drawn in advance.

One caution carries over from that earlier lesson: an alert without a response process is noise. Visibility creates value only when somebody is watching the signals, empowered and obliged to act on them — which is why the continuity organisation of the next section, not the software, is the heart of this strategy.

KEY CONCEPT

Flexibility: alternative corridors, alternative modes, postponement

The fourth strategy is flexibility: the designed ability to switch between ways of achieving the same result when the usual way closes. Where diversification answers who supplies, flexibility answers how goods move and when products take their final form.

1. Alternative corridors

A corridor is more than a road; it is a chain of route, port or border crossing, and clearance arrangements, as the lesson on International Transport Corridors mapped. Flexibility means knowing, before any disruption, which alternative corridor could carry the flow, what it would cost, how long it would take, and what must be arranged — documents, permits, partners — to make it usable.

2. Alternative modes and routes

Within a corridor, the lesson on Managing Multimodal and Cross-Border Shipments showed how transport modes combine. Resilience uses that knowledge under pressure: the critical line that normally moves by sea can move by air for the duration of a disruption, at a price that is only worth paying for the few lines whose stockout costs more than the freight. Modal flexibility is expensive precisely where it is valuable, and is reserved for the top of the critical list.

3. Postponement

The lesson on Demand Forecasting and Supply Planning taught postponement — keeping the product generic and delaying final configuration until real demand is known. In continuity terms, postponement is flexibility over demand itself: undyed yarn can serve whichever colour the market wants; unlabelled bulk can be finished for whichever market remains reachable. Delaying the decision keeps options open that early configuration would have closed.

Flexibility must be arranged before it is needed. An alternative corridor nobody has priced, a modal switch with no handling partner at the airport, a postponement decision never agreed with production — these are not options; they are ideas. Real flexibility is held open by

agreements, qualified partners and rehearsed decisions, all of which cost standing money, like every other currency resilience is bought in.

KEY CONCEPT

The continuity organisation: who does what when disruption strikes

Plans are executed by people, and a disruption begins with a question no software answers: who is in charge? The continuity organisation is the temporarily activated structure that runs the response — a deliberate assignment of roles, prepared in the plan, that turns a shaken group of employees into a functioning team.

Its core roles extend to organisational scale the incident roles of the lesson on Managing Shipment Delays, Damage, and Discrepancies.

1. The declarer

One named authority decides that a disruption exists and formally activates the plan. This is the highest-leverage decision in the response: too slow, and the alternatives expire unused; too eager, and the business pays disruption prices for ordinary noise.

2. The continuity lead

The coordinator of the response — the orchestrator role the lesson on Coordinating Carriers, Agents, and Regulatory Authorities described for a single move, now run across the whole operation. This role re-routes flows, activates fallbacks, reallocates stock and owns the recovery clock against the agreed recovery priorities.

3. The customer voice

One owner of external communication, so customers hear one consistent, honest account from one source. The earlier lesson's communication discipline applies unchanged: say what is known, what is not yet known, what is being done, and when the next update will come — then keep that promise.

4. The internal channel

Inside the organisation, a single source of truth replaces the corridor conversation: vacuums fill with rumour faster than any plan can, and staff who learn of a closure from social media do not make reliable decisions afterwards.

Communication also runs outward: suppliers must learn what is now expected, service providers re-instructed, and authorities reached through the proper channels. The test of the design is simple: within the first hour, everyone who needs to know something knows it, from the right source, and knows when they will hear more.

KEY CONCEPT

Testing, measuring, and renewing readiness

Resilience testing is the evaluation of an organisation's ability to respond to and recover from simulated or actual disruptions. It exists because of an uncomfortable truth: a plan that has never been rehearsed has never been read. Testing comes in increasing weights.

1. The tabletop exercise
The team sits around a table, a facilitator presents a simulated disruption — the clearance system is down, the primary corridor is closed — and the team talks its way through the response against the plan, without touching real cargo. Cheap, safe, and surprisingly exposing.

2. The drill
A single element is practised for real: the phone tree is called, the manual release procedure is run on one order, the alternative route is driven end to end.

3. The failover test
A system or process is genuinely switched to its fallback — the warehouse picks from the printed stock position for an afternoon — and the recovery time is measured against the recovery time objective.

Every test, and every real event, ends the same way: an after-action review asking what happened, what worked, what failed, and who changes what by when. The lesson on Managing Shipment Delays, Damage, and Discrepancies required every incident to end in a root cause and a change; continuity applies the same rule to the plan itself, so that events become design changes rather than anecdotes.

Readiness is then measured — the lesson on Logistics Performance Indicators and KPIs holds here as anywhere: people manage what is measured. Basic measures of continuity readiness include the share of critical activities with a documented, tested fallback; the date of each plan element's last test; the declaration-to-fallback time achieved in the last exercise; the currency of the contact tree and offline records; and recovery objectives defined for, and met in test by, every critical system. A rhythm of test, measure, fix and re-test keeps the plan alive — and is the subject of this lesson's closing warning.

WORKED EXAMPLE

The concentration audit: a Tema importer maps its single route

Consider an importer of industrial spare parts and machinery components based at Tema, supplying mines and manufacturers across Ghana. Its fastest-moving critical lines are sourced from a manufacturer in Asia. All company details and quantities in this example are illustrative for teaching purposes.

Step 1: Draw the concentration map
The logistics manager traces the inbound flow of the critical lines from purchase order to shelf:

one supplier in Asia; containers on one shipping line's service, routed via the Suez Canal; discharge at one terminal at Tema Port under one handling agreement; clearance through one agent; storage in one warehouse. Five circles on the map — supplier, corridor, terminal, agent, warehouse — each a single point of failure.

Step 2: Test the map against the record
The manager does not need to imagine a failure; the verified record supplies one. Attacks on shipping in the Red Sea led major container lines, from December 2023, to reroute widely around the Cape of Good Hope instead of the Suez Canal, substantially lengthening their voyages — and the service carrying this importer's containers was among them. Lead times stretched by weeks, and the importer changed nothing to cause it. Single-route assumptions fail on the world's schedule, not the importer's.

Step 3: Assess likelihood and impact
Qualitatively: corridor disruption, likelihood high in recent years, impact high, because the critical lines keep customers' equipment running. Terminal or agent failure, likelihood lower, impact still high. The pairings say the corridor deserves the first spending.

Step 4: Weigh the mitigations, trade-offs included
A strategic buffer of the most critical spare parts: carrying cost, but it covers the stretched weeks. A qualified air-freight option for the short list of lines whose stockout stops a customer's production: premium freight, reserved for that list, with a named person deciding when it is justified. A standing agreement with a second clearing agent: a modest duplication that erases one circle. Shared forecasts with the supplier and earlier ordering, so schedule changes arrive as warning instead of surprise.

Step 5: Write the one-page plan
The critical list with recovery priorities, the agreed buffer depths, the air-freight trigger and its decision-maker, the second agent's details, and a date next year to redraw the map.

What does this example demonstrate?
The audit converts unease into decisions. Every circle on the map became a named, priced, owned mitigation — chosen before the next disruption, not during it.

Key lesson
A concentration you have mapped, priced and covered is a managed risk. A concentration you have never drawn is a surprise with your name on it.

WORKED EXAMPLE

The tabletop exercise: half a day without the system

A pharmaceutical distributor in Accra clears most of its stock through the port and distributes nationwide from a single central warehouse. Its continuity plan, written a year earlier, has never been tested. The new operations manager schedules a tabletop exercise: half a day, the

management team, and a simulated outage of the electronic systems used for clearance declarations and warehouse stock records — rehearsed on paper, with no real stock touched.

Step 1: The scenario is announced
At 09:00 the facilitator reads the scenario: the clearance platform and the warehouse stock system are unavailable, expected to last half a day. Queued: a refrigerated consignment of temperature-sensitive medicines, fourteen customer orders, and a supplier asking whether to dispatch a scheduled consignment.

Step 2: The team walks the plan
The plan names the operations manager as declarer — done, on paper, within ten minutes. The team talks through the manual fallbacks: release processing on paper forms, picking against the previous day's printed stock position, the phone tree notifying the three largest customers, and the cold-chain supervisor giving the refrigerated consignment first claim.

Step 3: The gaps appear
Then the plan meets the table. The phone tree holds the numbers of two people who left the company last year. Nobody can say where yesterday's stock position would be printed from, because the printing routine died when the system arrived. No customer message template exists, so each of the fourteen calls would be improvised. And nothing in the plan answers the supplier's question — who may decide a dispatch that would arrive at an unresponsive warehouse?

Step 4: Fix, assign, and book the next test
The gaps become actions before anyone leaves the room: the phone tree is corrected and printed inside the plan; a close-of-business printed stock position becomes a standing daily routine, joining the warehouse's documentation discipline; two message templates are drafted; and the declarer's authority is extended to routine dispatch decisions. A ten-minute quarterly drill of the phone tree is scheduled, and the next tabletop is set for before the rainy season.

What does this example demonstrate?
A plan that read well in the office failed at the table — which is exactly what the table is for. Each gap fixed at the table cost minutes; the same gaps discovered during a real outage would have cost a cold chain, fourteen customer promises and the confidence behind them.

Key lesson
Where the lesson on Coordinating Carriers, Agents, and Regulatory Authorities rehearsed escalation for one live shipment, resilience testing rehearses the whole operation in calm, so that the real event finds no beginners.

WORKED EXAMPLE

Dual sourcing priced honestly: a Rotterdam manufacturer weighs two suppliers

A kitchen-equipment manufacturer in Rotterdam buys a critical electronic control module from a single supplier overseas. The unit is dependable, the price the best the team has found, and

the sea lead time long. A new supply chain manager draws the concentration map, circles the module, and proposes qualifying a second supplier nearer to home. All figures in this example are illustrative for teaching purposes.

Step 1: Establish the two positions
Supplier A, the incumbent: 100 US dollars per unit, a six-week sea lead time, one production site. Supplier B, the candidate: 110 US dollars per unit, a one-week road lead time, two production sites in different countries. Annual demand is 2,000 units, so any volume moved to B carries a premium of 10 US dollars per unit.

Step 2: Size the exposure honestly
If supplier A fails — a fire, an export restriction, a congested corridor — assembly stops. The manufacturer estimates one major outage: idle assembly labour, contractual late-delivery penalties, expediting, management time and sales lost for good, together plausibly 250,000 US dollars for a stoppage of several weeks. Against the record of recent years — pandemic-era congestion, a canal closure, mass rerouting — a one-in-ten chance of such an outage in a given year is not a pessimist's estimate. The expected annual cost of doing nothing is therefore around 25,000 US dollars, against which a premium of 20,000 US dollars a year is not obviously too high, and a smaller protected share is cheaper still.

Step 3: Structure the dual position
The company does not switch wholesale: supplier A keeps the primary share and its price advantage; supplier B receives a committed minority share — say a fifth of volume — enough to keep qualification, tooling and the relationship alive. Both pass the full evaluation discipline, and B's short lead time and second site become the recovery path if A fails.

Step 4: Apply the same logic per product family
For standard screws and brackets the answer differs: one supplier plus a modest buffer is cheaper, because a stockout there is fixable within days at trivial cost. Diversification is reserved for the short list of components whose failure stops the line.

What does this example demonstrate?
The premium is not a cost to be minimised but a price to be compared with an exposure. Judged only against the unit price, supplier B loses every time; judged against the risk it retires, supplier B is cheap.

Key lesson
Price resilience against the loss it prevents, not against zero. The honest comparison is never B's price against A's price; it is the premium against the outage.

IMPORTANT NOTE

Resilience is bought with money somewhere

Across every strategy in this lesson, resist the temptation of the free lunch: resilience is always paid for. The only real question is which currency the business prefers to pay in.

1. The premium
Diversification's currency: a higher price per unit, paid on every order, in exchange for a second way that exists.

2. The buffer
Stock's currency: cash converted into goods sitting still — capital tied up, space occupied, shelf life burning — paid continuously whether or not disruption ever comes.

3. The speed
Flexibility's currency: the expensive option taken late — air instead of sea, premium handling, spot prices in a tight market — paid rarely, in a lump, at the worst possible moment for negotiation.

Visibility and preparedness carry a fourth, quieter currency: attention — people watching, testing, meeting, keeping records current.

The lesson on Inventory Optimisation and Service Levels taught that holding cost and stockout cost pull against each other, and that the right answer is a deliberate point between them. Continuity is that same trade at strategic scale. A chain optimised purely for cost holds no slack and looks brilliantly efficient until the day it stops; a chain armoured against everything prices itself out of its market. The professional position is neither extreme: spend resilience money where the business impact analysis says the pain would concentrate — the critical lines and the circled dependencies — and accept exposure where an interruption is cheap and quick to fix.

IMPORTANT NOTE

The map and the plan decay

Two warnings close the lesson: both failures are silent, and both are certain without maintenance.

Concentration feels efficient until it fails. The savings of a single supplier, corridor, terminal or system accrue quietly every day and show up in every cost report, while the exposure accrues invisibly and is charged in full on one afternoon. This asymmetry is why concentration is never corrected by the accounts alone, and why the audit of single points of failure is an annual duty rather than a one-off project: suppliers are acquired, corridors close, providers merge, systems are replaced — and the map drawn last year no longer describes the business running today.

Plans decay faster than maps. People leave and take their telephone numbers; procedures change; the manual fallback that assumed paper forms dies quietly when a new system arrives, as this lesson's tabletop exercise showed. A continuity plan is not a document achieved but a capability maintained, and the maintenance is a rhythm in the same spirit as the review discipline of the lesson on Logistics Performance Indicators and KPIs: test on a schedule, measure readiness, fix what failed, re-test. When a real disruption ends, the after-action review feeds the plan, so that every event leaves the organisation measurably harder to disrupt than before.

Resilience, finally, is not a document or a warehouse or a second supplier. It is the habit of preparing on calm days for the days that will not be calm — kept alive by rhythm, or lost to quiet decay.

Practice questions (Lesson Checks)

Practice Question 1

A trading company in Accra imports all of its goods through a single overseas corridor, clears everything at one port terminal, and stores all stock in one warehouse. The finance manager opposes any spending on continuity arrangements, arguing that the cargo is fully insured, so the business is already protected against disruptions.

Which statement best describes the weakness in this reasoning?

1. Insurance compensates financial loss after an event, but it does not keep goods moving, customers served or the operation running, so the corridor, terminal and warehouse remain single points of failure carrying unmanaged risk. ✓

2. Insurance is unnecessary for trading companies, because well-run businesses can always find alternative suppliers and routes without preparing them in advance.

3. Disruptions are random and impossible to anticipate, so no preparation can improve the outcome and continuity spending is wasted in every case.

4. The company should immediately replace all of its single dependencies, whatever the cost, because any concentration at all is unacceptable in a professional supply chain.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

Insurance and continuity answer different questions. The lesson on Cargo Insurance and Risk Allocation showed that a policy transfers the financial consequence of a loss; it does not restore the flow of goods, honour customer promises or keep the warehouse operating while a claim is settled. The corridor, terminal and warehouse are single points of failure — dependencies whose failure stops the wider operation — and insurance leaves every one of them standing.

Answer 2 is the opposite error: alternatives only work if they were prepared in advance, qualified, priced and rehearsed; improvising them mid-crisis is the slowest and most expensive form of supply. Answer 3 is fatalism contradicted by the lesson's whole argument: disruptions may be impossible to schedule, but preparation reliably changes their consequences. Answer 4 over-corrects: resilience is bought deliberately where impact is high, and some concentration is accepted where exposure is cheap to fix.

The takeaway: insurance pays for what is lost; continuity planning keeps what is left running.

Practice Question 2

Ahead of the rainy season, the manager of a distribution business in Kumasi proposes four actions to the management team.

Which action is best described as contingency planning rather than risk prevention or disaster recovery?

1. Clearing the warehouse drainage and repairing the roof before the rains arrive, to make flooding less likely.

2. Documenting an alternative transport corridor and a second qualified supplier that can be activated if flooding closes the primary route. ✓

3. After a flood, pumping out the warehouse, drying the stock, and restoring the racking and systems to normal operation.

4. Paying the annual insurance premium under which flood damage to stored stock would be compensated.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Contingency planning is the preparation of alternative actions for responding to unexpected events. Like prevention, it happens before the event — but its aim is not to reduce the likelihood of the event; it is to have a ready answer when the event happens anyway. An alternative corridor and a second qualified supplier are exactly that: standing fallbacks that shorten the disruption instead of trying to prevent it.

Answer 1 is risk prevention, which acts on the likelihood or impact of the event itself. Answer 3 is disaster recovery — the restoration of facilities, systems and operations after the event has done its damage. Answer 4 is risk transfer, the layer examined in the lesson on Cargo Insurance and Risk Allocation: it compensates the loss financially but prepares no alternative action, so it is none of the three operational layers on its own.

The takeaway: prevention changes the odds, contingency prepares the alternative, recovery restores afterwards — and a mature continuity programme runs all three layers deliberately.

Practice Question 3

During a disruption, the ordering system of a wholesale business fails. Before the event, management had set two targets for this system: it must be restored and working again within two working days, and it may lose no more than the last four hours of recorded transactions.

Which option correctly identifies these two targets?

1. The two-day restoration target is the recovery time objective, and the four-hour data-loss limit is the recovery point objective. ✓

2. The two-day restoration target is the recovery point objective, and the four-hour data-loss limit is the recovery time objective.

3. Both targets are recovery time objectives, because both measure how quickly the system returns to normal service.

4. Both targets are service levels owed to customers under the supply contract, and they have no role in continuity management.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

The recovery time objective, or RTO, is the targeted maximum period within which a disrupted process or system should be restored — here, two working days. The recovery point objective, or RPO, is the maximum acceptable amount of data loss measured in time before a disruption — here, four hours of transactions. The RTO asks how fast the system returns; the RPO asks how much of its information survives. Together they turn recovery from a hope into a measurable promise that testing can verify.

Answer 2 swaps the two definitions: the time target measures the duration of recovery, not the loss of data. Answer 3 collapses the pair — speed of restoration says nothing about the transactions lost between the last record and the failure. Answer 4 confuses continuity measures with the customer-facing commitments of the previous lesson on Contract Management and Service-Level Agreements; both may exist in a business, but they are different instruments serving different promises.

The takeaway: the RTO is the clock on restoration, the RPO is the clock on data, and every critical system deserves both, in writing, before the disruption arrives.

Fact-check appendix (audit notes — not for entry)

The risk management cycle runs identify, assess (likelihood and impact), mitigate or treat, and monitor and review — source: research sheet logistics-scm-frameworks.md, section 11 (Risk and resilience).

The six risk categories used in this lesson (supply-side, transport, customs/regulatory, financial, security, natural) — source: research sheet logistics-scm-frameworks.md, section 11.

Business continuity planning as documented, tested plans for continuing or recovering operations, holding alternate suppliers, routes, sites and decision authority — source: research sheet logistics-scm-frameworks.md, section 11.

The resilience strategy set taught here (dual/multi-sourcing, buffer stock, end-to-end visibility, flexibility through alternative modes and routes, supplier development and collaboration) — source: research sheet logistics-scm-frameworks.md, section 11.

COVID-19 port congestion ran from early 2020 through roughly 2021–2022, involving demand shifts, labour disruptions, vessel and container imbalances, and record freight rates on major lanes; timing is kept broad and qualitative — source: research sheet logistics-scm-frameworks.md, section 11.

The Ever Given ran aground in the Suez Canal on 23 March 2021 and was freed on 29 March 2021, blocking the canal for six days and seven hours; at least 369 vessels were waiting by 28 March and over 400 by the reopening — source: research sheet logistics-scm-frameworks.md, section 11 (Wikipedia, "2021 Suez Canal obstruction").

Attacks on shipping in the Red Sea and Gulf of Aden began in October 2023, and from December 2023 major container lines widely rerouted around the Cape of Good Hope, substantially lengthening voyages between Asia and Europe; no percentage figures for voyage length or capacity are claimed — source: research sheet logistics-scm-frameworks.md, section 11 (Wikipedia, "Red Sea crisis").

Definitions of business continuity, business continuity plan, supply chain resilience, operational disruption, risk assessment, business impact analysis, contingency planning, disaster recovery, recovery time objective, recovery point objective, supplier diversification, redundancy, single point of failure and resilience testing follow the curriculum's key-term definitions for this lesson — source: lesson brief 15.3, Key Terms.

Postponement (delayed differentiation) keeps the product generic and defers final configuration until real demand is known — source: research sheet logistics-scm-frameworks.md, section 8 (Planning).

Safety stock as a buffer against demand and lead-time variability, holding-cost trade-offs, and segmented service levels — source: research sheet logistics-scm-frameworks.md, section 5 (Inventory management); lesson 13.3.

Tabletop exercises, drills, failover tests and after-action reviews as continuity testing practice; the insurance-premium framing of redundancy costs; the three-currency framing (premium, buffer, speed) — source: general business knowledge, consistent with the research sheet's resilience concepts.

Lesson 15.4 — Sustainability, Ethics, and Continuous Improvement

Summary

This lesson examines the importance of environmental responsibility, professional ethics, and continuous improvement in modern logistics management. Learners will explore sustainable transportation, energy efficiency, waste reduction, ethical procurement, regulatory compliance, and responsible business practices. The lesson also introduces structured improvement methods and explains how organisations can reduce environmental impacts while maintaining operational efficiency, accountability, and long-term competitiveness.

Learning objectives

- Define sustainable logistics and explain its importance.
- Identify major environmental impacts associated with transportation and warehousing.
- Describe measures for improving energy efficiency and reducing logistics emissions.

- Explain the importance of responsible waste management and resource utilisation.
- Identify ethical issues affecting procurement, transportation, and logistics services.
- Explain the importance of transparency, integrity, and regulatory compliance.
- Describe how continuous improvement supports operational performance.
- Identify common approaches used to analyse and improve logistics processes.
- Explain how sustainability performance can be measured.
- Recommend practical improvements that balance cost, service quality, environmental responsibility, and ethical considerations.

Key terms

Sustainable Logistics — The planning and management of logistics activities with consideration for environmental, social, and economic impacts.

Green Logistics — Logistics practices intended to reduce environmental harm associated with transportation, storage, packaging, and distribution.

Carbon Footprint — The total greenhouse gas emissions associated with an activity, organisation, product, or service, expressed in carbon dioxide equivalents.

Greenhouse Gas Emissions — The release of gases that contribute to the greenhouse effect, including carbon dioxide and methane.

Energy Efficiency — The achievement of a specified output or service using less energy.

Reverse Logistics — The movement of goods from downstream locations back through the supply chain for returns, repair, reuse, recycling, or disposal.

Circular Economy — An economic approach that seeks to retain the value of products and materials through reuse, repair, refurbishment, recycling, and other resource-efficient practices.

Ethical Procurement — The acquisition of goods and services in accordance with principles of fairness, integrity, transparency, and responsible conduct.

Corporate Social Responsibility (CSR) — An organisation's approach to managing its social, environmental, and ethical responsibilities.

Environmental, Social and Governance (ESG) — Categories used to assess environmental impacts, social responsibilities, and organisational governance practices.

Continuous Improvement — An ongoing process of identifying and implementing changes to improve operational performance.

Lean Management — A management approach focused on delivering value while identifying and reducing waste in processes.

Kaizen — A continuous improvement philosophy emphasising ongoing, incremental improvements.

Root Cause Analysis — A structured method for identifying the underlying causes of problems.

Plan-Do-Check-Act (PDCA) — A four-stage improvement cycle involving planning, implementation, evaluation, and corrective action.

Sustainability Indicator — A measurable value used to assess environmental, social, or economic sustainability performance.

Teaching content

KEY CONCEPT

Sustainable logistics defined, and why it pays twice

This lesson closes the programme with three disciplines that keep a logistics professional useful for a whole career: keeping the operation environmentally responsible, keeping it honest, and keeping it improving.

Sustainable logistics is the planning and management of logistics activities with consideration for environmental, social, and economic impacts. The definition asks a manager to weigh three consequences at once: environmental, social, and economic. Green logistics is the practical side of the same idea: logistics practices intended to reduce environmental harm associated with transportation, storage, packaging, and distribution.

Why does this belong in an advanced strategy module rather than in a corporate pamphlet? Because the environmental lens and the cost lens look at largely the same objects. Fuel is a cost, and burning it produces emissions; waste is a cost, and disposing of it burdens the environment; an empty vehicle is a cost, and it pollutes for nothing. Almost every measure that trims emissions trims expenditure with it, and Lesson 10.1, Components of Total Logistics Cost, already taught the arithmetic this extends: the sustainability lens and the cost lens mostly point down the same corridor.

Sustainability also belongs at strategy level because the big decisions are the big emitters. Lesson 15.1, Logistics Strategy and Service-Provider Selection, treated mode choice, network shape, and provider selection as the decisions that fix an operation's cost structure; they fix its environmental structure in the same stroke. A strategy that ignores these impacts is not neutral; it is a choice with one eye closed.

Where the two lenses genuinely part, a measure that costs money and buys no operational benefit, the decision is an ethical one, made deliberately, and this lesson treats it later.

KEY CONCEPT

Where logistics touches the environment: transport and warehousing

A logistics manager cannot manage impacts until the lesson names them. The major environmental impacts of the two great cost centres, transportation and warehousing, are these.

1. Exhaust emissions from transport
Vehicles, vessels, aircraft, and locomotives burn fuel to move freight, and burning fuel releases greenhouse gas emissions, the release of gases that contribute to the greenhouse effect, including carbon dioxide and methane. Transport exhaust also carries pollutants that affect local air quality along corridors and in port districts, so a fleet's impact is felt by neighbours as well as by the planet.

2. Energy consumption in warehousing
A warehouse buys energy continuously: lighting, ventilation, cooling, and, in the cold chain, refrigeration that must hold temperature around the clock. Handling equipment adds further demand. A warehouse's environmental signature is therefore mostly purchased energy.

3. Packaging waste
Protective packaging is often used for a single trip and discarded: cartons, wrap, void fill, pallets. Every tonne of packaging is material extracted, bought, used once, and disposed of.

4. Waste goods
Damaged, expired, obsolete, and returned stock all end as waste: the chilled consignment that failed its temperature record, the seasonal goods that never sold, the cosmetics past their shelf life. Each represents materials and logistics effort spent on goods that delivered no use.

5. Waste motion
The empty vehicle and the badly sited warehouse add movement that produces nothing, an impact examined among the levers shortly.

Two features of this list matter for management: every item is measurable, which the measurement concept later in this lesson exploits, and every item is also a cost line, which is why the environmental review and the cost review of an operation read like each other's translation.

KEY CONCEPT

The carbon footprint and the three scopes

To manage emissions, an organisation first frames them. The standard instrument is the carbon footprint: the total greenhouse gas emissions associated with an activity, organisation, product, or service, expressed in carbon dioxide equivalents. Carbon dioxide equivalents exist so that different gases can be added honestly: each gas is converted to the amount of carbon dioxide that would produce the same warming effect, and the totals are summed.

The most widely used accounting framework for organising that total is the Greenhouse Gas Protocol, developed by the World Resources Institute and the World Business Council for Sustainable Development. It splits the footprint into three scopes, and the split is easy to teach.

1. Scope 1: what you burn yourself
Direct emissions from sources the organisation owns or controls. For a fleet operator this is the dominant scope: the fuel its own trucks and forklifts burn.

2. Scope 2: the energy you buy
Indirect emissions from purchased electricity, steam, heat, and cooling. A warehouse that generates nothing itself still owns the emissions created far away to power its lights and its cold rooms.

3. Scope 3: everyone else in your value chain
Other indirect emissions across the value chain, including purchased transport. The freight a forwarder buys from airlines and shipping lines sits here, as do suppliers' and customers' operations.

The split matters because it disciplines two common self-deceptions. The first is claiming greenness by outsourcing: emissions do not vanish when the work is contracted out, they move to Scope 3, which is why a serious footprint counts the freight it buys, not only the vehicles it owns. The second is measuring only what is easy: for an asset-light logistics provider the owned scopes are small and the purchased carriage is enormous, so its carrier selection, the decision Lesson 15.1 formalised, is its largest emissions decision.

KEY CONCEPT

Lever one: modal shift and the honest ranking of modes

The single most powerful emissions lever in freight is modal shift: moving consignments from a higher-emitting transport mode to a lower-emitting one where the service promise allows it.

The direction of the shift rests on one of the few environmental claims logistics can state with confidence. Per tonne-kilometre of freight moved, sea is the lowest-emitting of the four main modes, rail is next, road is higher still, and air is by far the highest. Two honest qualifications accompany that ordering. The size of the gaps varies with the vessel, the vehicle, the load carried, the fuel, and the method of measurement, so no responsible text quotes fixed per-mode factors; but the direction of the ordering does not vary, because it follows the fuel burned to move a tonne, and lifting freight into the sky costs vastly more energy than floating it across an ocean.

Notice how little this changes the mode-selection framework of Lesson 3.2, Modes of Transport: Sea, Air, Road, Rail, and Multimodal. That lesson taught mode choice as a trade among speed, cost, capability, and availability. The environmental question rarely reverses that trade; it strengthens the slower, cheaper side of it, because the modes that charge less per tonne-kilometre are largely the modes that emit less. Modal shift therefore usually takes the form of

segmentation: the regular, forecastable base flow of replenishment stock moves to sea or rail where a longer lead time is acceptable, and the fast mode is reserved for the genuinely urgent slice of demand that justifies its premium on every dimension, including emissions.

Availability remains a real constraint: rail requires a corridor, sea requires a port and time. Modal shift is a lever to pull where geography and the service promise permit, as the worked example later in this lesson shows in full.

KEY CONCEPT

Levers two to five: fuller loads, no empty kilometres, leaner packaging, efficient buildings

Lever one, modal shift, filled the previous concept. The remaining four operate inside the mode already chosen, in the carton, and in the building, and each was met earlier in this programme as a cost discipline before returning here as an environmental one.

2. Load-factor improvement

Load factor is the share of available capacity that actually carries cargo, the vehicle capacity utilisation measured in Lesson 13.4, Logistics Performance Indicators and KPIs. A half-empty truck covering a full truck's distance burns most of a full truck's fuel to deliver half the freight, so raising load factor cuts emissions per tonne-kilometre without changing anything else. The instruments are consolidation's: combining orders into fewer, fuller loads, the groupage and hub logic of Lesson 13.2, Transport Route and Network Optimisation. What consolidation does for cost per pallet it also does for emissions per pallet.

3. Empty-run reduction

Lesson 13.2 defined empty running as a vehicle travelling without paying cargo, and taught backhaul, the return leg ideally carrying paying cargo, as its cure. The environmental restatement is stark: an empty leg produces emissions with no freight to show for them, the purest waste in transport. Beside these sits route and network efficiency, planning fewer and better-chosen kilometres, also from Lesson 13.2; the greenest kilometre remains the one not driven.

4. Packaging optimisation

Packaging optimisation achieves the same protection of the goods with less material and less lost space: right-sized cartons that fit the product instead of surrounding it with void fill, lighter materials where strength allows, and reusable transit packaging such as returnable pallets and crates where flows allow. The benefits compound: less material is bought and disposed of, and smaller, denser cartons mean more cubes filled per vehicle and per container, which feeds the load-factor lever directly.

5. Energy efficiency in the warehouse

Energy efficiency is the achievement of a specified output or service using less energy. The definition names the honest way to improve: the same output from less energy. A warehouse pursues it on familiar fronts, lighting and equipment that run where work is happening rather

than everywhere always, cold rooms whose doors and seals keep cooled air where it is paid to be, and space used efficiently so less building is lit, cooled, and ventilated for the same stock.

All four pay three times at once, in cost, in emissions, and usually in simpler, more reliable service.

KEY CONCEPT

Reverse logistics and the circular economy: waste as misplaced resource

Sustainable logistics treats waste not as an unavoidable residue but as resources in the wrong place. Two concepts carry that view.

Reverse logistics is the movement of goods from downstream locations back through the supply chain for returns, repair, reuse, recycling, or disposal. Lesson 1.4, Types of Logistics: Inbound, Outbound, Reverse, and Third-Party, introduced it as the third flow beside inbound and outbound; here it returns as environmental infrastructure. Every returned pallet, reusable crate, refurbishable device, and recyclable bale of packaging travels backwards through exactly the transport, handling, and records discipline the programme has taught throughout.

The circular economy is the economic approach that seeks to retain the value of products and materials through reuse, repair, refurbishment, recycling, and other resource-efficient practices. Applied to a logistics operation, it produces a simple ordering of responses to an end-of-life item: prefer the outcome that retains the most value, reuse before repair, repair before recycling, recycling before disposal. Each step down the ordering destroys value that the step above would have kept.

Responsible waste management in a warehouse follows from this. Waste streams are segregated so that recyclable material is not contaminated into worthlessness; packaging and damaged goods are recovered where recovery pays; and what must be disposed of is disposed of through proper channels. Two further duties sit with the logistics professional. The first is honesty: written-off and expired goods leave the records cleanly, per the documentation discipline of Lesson 8.4, Warehouse Documentation and Stock Reconciliation, and never quietly re-enter any market, a duty that protects the public as much as the environment. The second is arithmetic: waste is material the business paid for twice, once to buy and once to discard, so the waste review is a cost review wearing different clothes.

KEY CONCEPT

Measuring sustainability performance honestly

A sustainability indicator is a measurable value used to assess environmental, social, or economic sustainability performance. The definition is deliberately broad, and the breadth is honest: no operation measures everything, and none needs to. Build the working set from what the records can actually support.

1. Fuel-based indicators
Litres of fuel consumed per delivery, or per tonne-kilometre, drawn from fuel issues and trip records. It is the workhorse transport indicator, improved by every lever of this lesson.

2. Mode and utilisation indicators
The share of tonne-kilometres moved by each mode, and the empty-running share of kilometres run. Both report directly on the modal shift and empty-run levers.

3. Energy and waste indicators
Electricity consumed per period against throughput for the warehouse; packaging weight per order shipped; waste recycled or reused as a share of total waste generated.

4. Trend discipline
Every one of these is read as a trend against the operation's own baseline, exactly as Lesson 13.4 taught: formula stated, data source fixed, owner named, baseline established before targets, definitions never quietly changed.

Honesty deserves its own paragraph, because sustainability measurement attracts more decorated numbers than almost any field. Perfect carbon accounting, a complete footprint down to the last kilogram, lies beyond most operating companies, and importing other organisations' footprint figures as benchmarks repeats exactly the error Lesson 13.4 dismantled: numbers produced under other conditions are not comparable. The professional standard is different and achievable: measure what you can, trend it honestly, and use the verified directional knowledge of this lesson, the mode ranking above all, to point the trend in the right direction. A fleet whose fuel per tonne-kilometre falls year on year is measurably improving, even if nobody in the business can state its exact tonnage of carbon dioxide equivalents.

KEY CONCEPT

Ethical procurement, CSR, and ESG: integrity beyond the border

Sustainability has a social and governance face as well as an environmental one, and in logistics it shows first in buying.

Ethical procurement is the acquisition of goods and services in accordance with principles of fairness, integrity, transparency, and responsible conduct. Freight and logistics services are bought at scale, so procurement's integrity risks are operational risks. The familiar ones include undisclosed conflicts of interest, buying from a business a colleague quietly owns; kickbacks and hospitality that steer an award; favouritism that skips competition; and dishonest specification, requirements written around a preferred supplier. The defences are procedural, and this programme has already supplied them. Lesson 15.1, Logistics Strategy and Service-Provider Selection, taught specification first, written criteria, and price evaluated last: the selection discipline is an integrity device as much as a quality one, because a decision made against published criteria leaves little room for favour. Lesson 15.2, Contract Management and Service-Level Agreements, taught governance after signature: fair dealing continues through the

contract's life, with partners paid on agreed terms and performance judged on evidence, not manufactured.

Fair treatment also runs downward and outward: drivers given safe vehicles and legal working hours, staff not pressured to cut corners, partners' people treated as the professional would wish to be treated.

Two umbrella terms frame these duties from outside. Corporate social responsibility, CSR, is an organisation's approach to managing its social, environmental, and ethical responsibilities. Environmental, social and governance, ESG, refers to the categories used to assess environmental impacts, social responsibilities, and organisational governance practices. Their practical relevance is concrete: large customers, lenders, and partners increasingly ask structured questions about all three, and a provider's answers are only as good as its actual conduct.

KEY CONCEPT

Facilitation payments, bribes, and the transparency test

Freight and customs run on official decisions: releases, inspections, permits, clearances. That is precisely why the trade carries a characteristic integrity risk, met operationally in Lesson 11.3, Supply Chain Security and Cargo Integrity, and at the border in Lesson 12.2, Transit Logistics and Border Operations.

The risk turns on a distinction every professional must be able to state. A facilitation payment is a payment to expedite a routine, non-discretionary government action, one whose outcome is already fixed and whose only variable is speed: the documents are in order, the goods are as declared, and the decision will be the same whenever it arrives. A bribe, by contrast, is intended to change the decision, to release goods that should be held, pass goods that should be inspected, reduce a value that should stand. The two sit on opposite sides of a line, but the line is narrower than it looks, and the law does not treat it uniformly. Some jurisdictions carve out a narrow exception for payments covering routine governmental action; others, such as the United Kingdom under its Bribery Act 2010, recognise no exemption and can treat facilitation payments as unlawful. Many multinational companies' codes of conduct close the argument by prohibiting both bribes and facilitation payments. The professional rule follows: follow the stricter standard that governs you, normally the company code, because permitted somewhere is not the test.

Document integrity is the same commitment in ink. Lesson 6.1, Commercial Invoice and Packing List, allows no second invoice and no reused number; Lesson 7.2, Air Cargo Operations and Cargo Acceptance, reduced signing to one habit: never sign a cargo description you cannot stand over. False statements of origin, value, or contents are where freight fraud begins, and the innocent shipper answers for what is found in its consignment.

The practical test that binds it all is transparency: would this decision survive being written down and read aloud, to a customer, an auditor, a colleague? A decision that would is defensible. A decision that would not has already told you what it is.

KEY CONCEPT

Continuous improvement: the PDCA cycle and lean thinking

The third closing discipline is continuous improvement: an ongoing process of identifying and implementing changes to improve operational performance. The definition's weight falls on ongoing. A logistics operation generates errors, delays, and waste every single day; an occasional improvement effort cannot win a contest against decay that runs always.

The classic method is the plan-do-check-act cycle, PDCA, associated with the quality statisticians Shewhart and Deming, and its four stages are a complete instruction.

1. Plan
Measure the process, identify the biggest problem and its cause, and plan a change small enough to trial.
2. Do
Implement the change on a limited scale, one line, one zone, one lane.
3. Check
Measure again, against the baseline, and compare honestly.
4. Act
Standardise what worked so it becomes the new normal, adjust what did not, and begin the next cycle.

Lean management supplies the map of where to hunt. It is a management approach focused on delivering value while identifying and reducing waste in processes, and its catalogue of wastes includes overproduction, waiting, unnecessary transport, excess inventory, defects, over-processing, and unused talent. This programme has walked that catalogue operationally: the travel waste of Lesson 8.1, Warehouse Functions and Layout; the empty running of Lesson 13.2; the defect chains of the warehouse process lessons. Lean's claim, which the cost lessons proved, is that waste hides inside daily routine until someone measures and names it.

One tool disciplines the whole cycle: root cause analysis, a structured method for identifying the underlying causes of problems, taught in Lesson 11.4, Managing Shipment Delays, Damage, and Discrepancies, whose rule was blunt: every incident ends with a root cause and a change, or it will repeat. Fixing the symptom, retraining everybody, re-counting everything, leaves the cause in place to regenerate the problem. Continuous improvement supports operational performance for this compounding reason: small causes produce daily losses, and small permanent fixes compound daily against them.

KEY CONCEPT

5S, kaizen, and the rhythm that closes the book

Two further disciplines translate improvement from method into workplace culture.

5S is workplace organisation practised in five steps: Sort, remove from the work area what the work does not need; Set in order, give what remains a labelled, logical home; Shine, clean and inspect, because cleaning is where leaks, wear, and damage are noticed; Standardise, write the first three steps into the normal routine; and Sustain, keep to the standard when the novelty has passed. Kaizen is a continuous improvement philosophy emphasising ongoing, incremental improvements, involving everyone in small, frequent changes that are standardised when they work. Their shared insight is that an organised workplace makes abnormalities visible and improvement cheap: in a sorted, ordered warehouse, a misplaced pallet announces itself.

Around the tools runs the rhythm this programme has been assembling lesson by lesson:

1. Measure, with the KPI discipline of Lesson 13.4.
2. Find the biggest waste, with lean's catalogue as the search map.
3. Fix the cause, not the symptom, with root cause analysis.
4. Standardise, so the fix survives staff rotation and busy seasons.
5. Repeat, on the next-biggest cause, indefinitely.

And here the whole book closes, because every module supplied a part of that loop. Lesson 8.4's records make measurement possible; Lesson 13.4's indicators make waste visible; Lesson 11.4's incident discipline turns each failure into a cause found and a change made; Lesson 12.3's coordination spreads a fix across carriers, agents, and authorities; Lesson 15.3's resilience planning keeps the operation improving through disruption, not merely surviving it. Sustainability sets the direction, ethics sets the boundary, and continuous improvement is the engine that moves along both, one measured cycle at a time, for the length of a career.

WORKED EXAMPLE

The modal-shift decision: a Tema cosmetics exporter rebalances its lanes

A manufacturer of packaged shea butter cosmetics based in Tema exports twice monthly to a distributing partner in Rotterdam. Every consignment flies from Kotoka International Airport to Amsterdam, trucked the short final leg to Rotterdam. All figures in this example are illustrative for teaching purposes.

Step 1: Examine the current pattern
Each air consignment costs about 4,000 US dollars and reaches the partner's warehouse in roughly one week. The partner's sales are steady and forecastable; the air freight premium is embedded in the product's landed cost; and the entire intercontinental flow sits on the highest-emitting mode per tonne-kilometre.

Step 2: Segment the demand
Applying the segmentation logic of the modal-shift concept, the logistics manager asks the partner one question: which volume truly needs air? The answer: the base replenishment flow, roughly three quarters of volume, serves steady demand and can accept a longer lead time; only launch stock and urgent top-ups are time-critical.

Step 3: Compare the options honestly
Sea freight via Tema Port to Rotterdam costs about 1,500 US dollars per equivalent consignment and takes around three weeks port to port. The lesson's directional ranking applies: the same tonnes move from the highest-emitting mode to the lowest, so the emissions trend improves in direction, though nobody quotes a precise percentage, because none can be honestly known.

Step 4: Weigh the inventory consequence
The longer transit is not free, and Lesson 13.3, Inventory Optimisation and Service Levels, supplies the cost: goods at sea for around three weeks are pipeline stock, so the partner must hold a higher base level of inventory to cover the longer replenishment cycle. The parties set the carrying cost of the additional pipeline stock against the freight saving; the saving clearly dominates for this dense, non-perishable, stable-demand product, but the check is made explicitly, which is the discipline.

Step 5: Decide, then measure
The parties split the flow: replenishment base stock moves to sea on a scheduled monthly sailing, launch and urgent consignments stay on air, and the partner's ordering calendar is adjusted to the longer sea cycle. The exporter adds two indicators in the Lesson 13.4 format: freight cost per consignment by mode, and the share of tonne-kilometres moved by sea, both standing items in the monthly review.

What does this example demonstrate?
One structured decision served all three objectives at once: freight cost fell substantially, the emissions trend turned towards the lowest-emitting mode, and service was protected by segmenting rather than by moving everything. The inventory impact was weighed, not ignored.

Key lesson
Modal shift succeeds as segmentation: move the volume that can wait, keep the fast mode for what cannot, and let the trend lines record the improvement.

WORKED EXAMPLE

The hint at the border: an integrity dilemma handled at Aflao

A Ghanaian trading company runs a refrigerated truck carrying frozen fish from Tema to a buyer beyond the eastern border. At the Aflao border post, the queue for inspection is long, the afternoon is hot, and the cold-chain clock is running. All circumstances in this example are illustrative.

Step 1: The situation
The consignment's documents are complete and in order: the declaration, the health certificates, the transit paperwork all check out, prepared under the discipline of Lesson 12.2's early-working rule. The inspection itself is routine and non-discretionary: the goods and papers are in order, and the only variable is time.

Step 2: The pressure
A person at the post hints quietly that an informal payment would speed things up. The driver faces the classic compound pressure: the cargo is perishable, waiting has a real cost, the buyer is watching, and the hint implies that this is simply how things are done. The company's code of conduct, however, expressly prohibits both bribes and facilitation payments.

Step 3: Analyse what is being asked
The driver reasons it through exactly as the facilitation-payment concept teaches. The payment would expedite a routine action, changing timing and not the decision; it is the narrower case, not a bribe. But the analysis does not end there. The company code prohibits the narrower case too, and some jurisdictions treat it as unlawful. Paying would also have to be recorded, and recording it as an approved border cost would conceal what actually happened: a false record is a second wrong stacked on the first.

Step 4: Respond and escalate
The driver declines courteously and follows the procedure: he contacts the transport supervisor, who informs the company's compliance focal point. The event is recorded factually, time, post, what was hinted, no embellishment. The shipment waits its turn and clears through the proper channel some hours later; the cold chain holds, and the buyer is informed honestly of the delay.

Step 5: Learn from the event
The review afterwards asks the improvement question: was anything in the company's own preparation adding to the border delay? Document sequencing is tightened and the pre-departure checklist gains a line, so future crossings arrive even cleaner.

What does this example demonstrate?
The professional response to an integrity probe is the same under heat as on paper: refuse courteously, escalate through the declared channel, record factually, never launder the event through the cost records, and let the process close the loop.

Key lesson
Ethics has no cost-benefit test. The frozen cargo, the long queue, and the watching buyer change the pressure, never the answer.

WORKED EXAMPLE

The kaizen quarter: one PDCA cycle on pick errors in Rotterdam

A distribution centre near Rotterdam serving e-commerce customers picks thousands of order lines a week. Management wants fewer picking errors, and the continuous improvement

manager proposes one plan-do-check-act cycle on the single biggest error cause. All figures in this example are illustrative for teaching purposes.

Step 1: Plan, from measurement
Following the Lesson 13.4 discipline, the team first establishes a baseline: order accuracy records and error logs for a full month show 100 picking errors, each logged with its order, item, and location when discovered. Root cause analysis, sorting the errors by underlying cause rather than by blame, finds that 40 of the 100 trace to one cause: two pairs of look-alike products, near-identical packaging with different pack sizes, stored in adjacent locations. The planned change is small and targeted: separate the look-alike items to non-adjacent zones and require a barcode scan confirmation at the picking locations concerned.

Step 2: Do, on a small scale
The change is trialled for the affected items only, in one aisle, for two weeks. Pickers are shown the new locations and the scan step; nothing else in the building changes.

Step 3: Check, against the baseline
Over the following month the team re-measures. Errors from the look-alike families fall from 40 to 8; total errors fall from 100 to 62. The improvement is real, concentrated where the change was made, with no rise in picking time worth noting.

Step 4: Act, by standardising
What worked becomes the standard. The slotting rule is rewritten, look-alike products are never stored in adjacent locations, and scan confirmation is extended to all picking locations, not only the trialled aisle. The revised standard goes into the warehouse's operating documentation, and the check remains a standing item in the weekly review, so a future relapse is seen.

Step 5: Repeat, on the next cause
The cycle restarts with the remaining 62 errors and their now-biggest cause. A quarter holds room for two or three such cycles.

What does this example demonstrate?
One complete PDCA cycle, run on a measured cause with a small trial and an honest re-measurement, converted a vague ambition, fewer errors, into a permanent standard and a falling trend, without a campaign, a poster, or a single blame.

Key lesson
Measure first, fix the cause, trial small, standardise what survives the check, and repeat: the rhythm matters more than the size of any single fix.

IMPORTANT NOTE

Sustainability fails when it ignores cost and service: keep the three in balance

The commonest way a sustainability initiative dies is not resistance to the environment; it is collateral damage to cost or service. This lesson's levers were presented with their commercial

face showing, and the presentation was deliberate, but the balance still has to be managed case by case.

The failure pattern is recognisable. A company bans air freight outright and discovers that its urgent, launch, and high-value consignments now miss their promises, and Lesson 1.3, The Seven Rights of Logistics, has already ruled on a delivery that arrives late: however green, it is a service failure. A distribution business pushes consolidation to its maximum and re-learns Lesson 13.4's tension trap: fuller trucks, slower promises, complaining customers. A warehouse cuts lighting to a minimum and finds picking accuracy, and with it order accuracy, quietly deteriorating. Each initiative improved the environmental number by spending the service or the cost account it forgot to watch.

The professional recommendation therefore always names all four accounts: cost, service quality, environmental responsibility, and ethical considerations. The modal-shift worked example is the model: the exporter did not abandon air, it segmented, moved what could wait, kept the fast mode for what could not, weighed the inventory consequence, and let the measurement report the balance.

The test before adopting any green initiative is consequently a pair of questions. Does it hold the service promise? And does it live within a cost position the business can sustain? An initiative that passes both is an improvement regardless of its environmental yield. An initiative that fails either will be abandoned within a year, and the abandonment will discredit the next three ideas behind it.

IMPORTANT NOTE

Ethics has no cost-benefit test, and no entry in the expense ledger

Cost-benefit reasoning is the daily bread of logistics, and this programme has taught it freely. This note exists to mark the boundary of that reasoning, because ethics is the one account where the arithmetic does not apply.

The rationalisations arrive in a predictable order, and each has a short answer. It is only a small payment: size has never been the test; the bribery distinction turns on what the payment is for, and the company code governs the facilitation case outright. It is customary here: custom describes what some people do, not what a professional may do, and many multinational codes prohibit both bribes and facilitation payments precisely because they decline to localise integrity. It only affects timing, not the decision: true as far as it goes, and insufficient, because the stricter standard still prohibits it, and in some jurisdictions the law does as well. Everyone does it: the transparency test answers this one alone, would the payment survive being written down and read aloud? Nobody advances that answer aloud.

Two companion rules complete the boundary. First, a wrong payment is never repaired by a wrong record: booking a facilitation payment as a normal border or freight expense launders the event into the accounts, and the document integrity of Lesson 6.1 and the reconciliation

discipline of Lesson 8.4 forbid records that misstate what happened. Second, escalation protects the professional as much as the organisation: the driver who reports the hint at the border has documented his own integrity at the moment it was tested.

Regulatory compliance is the floor beneath all of this, not the ceiling. A professional who acts only as enforcement probability permits pays oversized premiums on every bad day; one who holds the line as a matter of identity pays none.

IMPORTANT NOTE

Continuous improvement is culture, not campaign

The final warning concerns pace and persistence. Improvement programmes fail more often from grandeur than from difficulty: the launch event, the poster campaign, the quarter of focus, then the quiet return to routine while unfixed causes resume their daily work. Continuous improvement is culture, not campaign: small, steady, and measured beats grand and abandoned.

Three habits distinguish the cultures that sustain the rhythm from the campaigns that flare and die.

1. They measure before and after
Every improvement claim in a measured culture is a before-and-after comparison against the operation's own baseline, in the Lesson 13.4 format. The same honesty governs sustainability claims: measure what you can, trend it, decline to import impressive numbers produced under other conditions.

2. They give every cycle an owner and a standard
A fix without a name beside it is a hope. The PDCA act stage ends with the change written into the standard, the slotting rule, the checklist, the border routine, and a person answerable for keeping it there, as Lesson 13.4 refused to let a KPI travel without its owner.

3. They run small cycles indefinitely
A quarter holds room for two or three complete plan-do-check-act cycles on the biggest current causes. Small cycles are affordable, reversible, and honest; a transformation programme is none of these, and its failure poisons appetite for the small next step that would have worked.

The closing thought is the loop itself. Sustainability sets the direction, ethics draws the boundary no result may cross, and continuous improvement provides the engine, fed by the records, measurements, incident lessons, coordination habits, and resilience plans the programme has built. A logistics professional who keeps that loop turning, one measured cycle at a time, will still be improving the operation long after this course is a certificate in a drawer.

Practice questions (Lesson Checks)

Practice Question 1

An exporter in Tema ships packaged cosmetics to a buyer in Rotterdam twice a month, entirely by air. The buyer confirms that its regular replenishment stock can accept a longer lead time, provided that urgent consignments still arrive quickly. Management wants to reduce freight cost and environmental impact without harming service.

Which option best describes the soundest response?

1. Move everything to smaller and more frequent air consignments, so that each shipment carries less and the impact of any single flight is reduced.

2. Shift the regular replenishment volume to sea freight, keep air freight for urgent and launch consignments, and track cost and emissions trends over time. ✓

3. Continue using air freight for everything, because air is the lowest-emitting mode per tonne-kilometre of the main transport modes.

4. Switch all volume to sea freight and accept that urgent consignments must now wait several weeks like everything else.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

This is the modal-shift decision as the lesson teaches it: per tonne-kilometre, sea is the lowest-emitting main mode and air by far the highest, so moving the volume that can wait turns the emissions trend in the right direction while cutting freight cost substantially. Segmenting the flow, sea for the forecastable base, air for what is genuinely urgent, protects the service promise, and tracking the trends applies the measurement discipline of Lesson 13.4.

Answer 3 rests on an inverted fact: air is the highest-emitting mode per tonne-kilometre, not the lowest. Answer 1 worsens every account at once, because smaller, more frequent consignments lower the load factor, raise cost per unit, and leave the flow on the highest-emitting mode. Answer 4 passes the environmental test by failing the service one: a consignment that arrives weeks late is a service failure however it travels, and the lesson's worked example showed the segmentation alternative instead.

The takeaway: modal shift succeeds as a segmented decision, weighed against lead time and inventory, and recorded as a trend.

Practice Question 2

A refrigerated truck carrying frozen fish waits in a long inspection queue at a border post. The consignment's documents are complete and the inspection is routine. An official hints that an

informal payment would speed things up. The driver's company code of conduct prohibits both bribes and facilitation payments.

Which response aligns with the lesson?

1. Make the payment and record it in the trip expenses as a normal border cost, since the perishable cargo justifies protecting the cold chain.

2. Make the payment without concern, because it would only change the timing of a routine action and not the decision itself.

3. Decline courteously, escalate through the company's declared procedure, record the event factually, and let the clearance follow the proper channel. ✓

4. Tell the driver to abandon the declared route and use a quieter crossing where no inspection queue is likely.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The response follows the professional sequence taught in the border worked example: refuse courteously, escalate through the declared channel, record the event factually, and let the routine inspection run its course. The facilitation-payment distinction does not license the payment here, because many company codes, including this one, prohibit both bribes and facilitation payments, and some jurisdictions treat facilitation payments as unlawful; the stricter standard that governs the professional is the company code.

Answer 1 adds a second wrong to the first: recording the payment as an approved border cost launders the event into the accounts, and the document integrity discipline forbids records that misstate what happened. Answer 2 is the timing-versus-decision rationalisation, which describes the narrow case correctly but draws the wrong conclusion, because the narrower case is exactly what many codes prohibit. Answer 4 evades the record entirely and leaves the integrity event unreported, so the company learns nothing and the driver's conduct is undocumented.

The takeaway: ethics has no cost-benefit test, and the perishable cargo changes the pressure, never the answer.

Practice Question 3

The manager of a distribution centre wants to reduce picking errors and asks how to apply the plan-do-check-act cycle.

Which sequence best applies PDCA as the lesson teaches it?

1. Launch an awareness campaign with posters and meetings across all departments, then review overall performance after a year.

2. Measure the errors to identify the biggest underlying cause, trial a small corrective change, re-measure against the baseline after an agreed period, then standardise what worked and repeat on the next cause. ✓

3. Retrain all pickers immediately on general accuracy, then measure the error rate once at the end of the year.

4. Wait for the annual stock count to reveal the size of the problem, then commission a consultancy study before acting.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

This is the complete cycle: plan from measurement and root cause analysis, do on a small scale, check against the baseline, act by standardising what worked, and repeat on the next-biggest cause. The Rotterdam worked example ran exactly this sequence on look-alike products stored adjacently and turned 100 monthly errors into a falling trend and a rewritten slotting standard.

Answer 1 is the campaign pattern this lesson's final note warns against: broad, unmeasured, and abandoned when the posters fade, with no cause ever fixed. Answer 3 treats a symptom and skips the diagnosis: retraining everyone assumes the cause is knowledge, where the measured cause was storage layout, and a single end-of-year measurement provides no baseline comparison along the way. Answer 4 has no rhythm at all: an annual count cannot locate picking causes, and waiting for an external study surrenders the improvement to delay.

The takeaway: continuous improvement is culture, not campaign, small measured cycles, standardised and repeated, outlast every grand launch.

Fact-check appendix (audit notes — not for entry)

Definitions of sustainable logistics, green logistics, carbon footprint, greenhouse gas emissions, energy efficiency, reverse logistics, circular economy, ethical procurement, corporate social responsibility, environmental, social and governance, continuous improvement, lean management, kaizen, root cause analysis, plan-do-check-act, and sustainability indicator follow the lesson brief 15.4 key terms. — source: lesson brief 15.4, Key Terms (curriculum database).

Green logistics reduces logistics' environmental impact (emissions, energy, waste) via cleaner modes, load optimisation and packaging reduction; reverse logistics is the flow of goods back toward origin (returns, repairs, recycling, disposal) and sits inside logistics management. — source: research sheet logistics-scm-frameworks.md, section 1.

GHG Protocol (developed by WRI and WBCSD): Scope 1 = direct emissions from sources the company owns or controls; Scope 2 = indirect emissions from purchased electricity, steam, heat, cooling; Scope 3 = other indirect

emissions across the value chain, including purchased transport. — source: research sheet logistics-scm-frameworks.md, section 12 (GHG Protocol Corporate Standard).

Per tonne-kilometre, the directional emissions ranking of the main modes is sea < rail < road < air; the lesson teaches the ordering only, with the explicit qualification that gap sizes vary by vehicle, load factor, fuel and measurement method, and quotes no emissions factors. — source: research sheet logistics-scm-frameworks.md, section 12 and DO NOT CLAIM item 1.

The recognised sustainability levers taught are modal shift to lower-intensity modes, load-factor improvement, empty-run reduction, packaging optimisation (lighter, right-sized, reusable) and reverse logistics for returns/refurbishment/recycling; empty running (travelling without paying cargo) is a major cost and emissions inefficiency, reduced via backhaul matching and network design. — source: research sheet logistics-scm-frameworks.md, sections 6 and 12.

A facilitation ("grease") payment expedites a routine, non-discretionary action whose outcome is already fixed, changing timing rather than the decision; the US FCPA treats payments for routine governmental action as a narrow exception, the UK Bribery Act 2010 recognises no exemption, and many multinational codes prohibit both bribes and facilitation payments; facilitation payments versus bribes is a core integrity risk in freight and customs. — source: research sheet logistics-scm-frameworks.md, section 13 and DO NOT CLAIM item 17 (neither "legal everywhere" nor "bribes under all laws" is claimed).

Trade-document manipulation, false origin/value declarations and misuse of delivery terms create fraud and compliance exposure; controls include document checks, three-way match and approved carrier/broker lists (taught generically, no specific allegations). — source: research sheet logistics-scm-frameworks.md, section 13.

PDCA (plan-do-check-act) is the Deming/Shewhart cycle: plan a change, do it small-scale, check results, act to standardise or adjust, repeat. Lean thinking maximises customer value by systematically eliminating waste, including overproduction, waiting, unnecessary transport, excess inventory, defects, over-processing and unused talent. 5S is workplace organisation in five steps (Sort, Set in order, Shine, Standardise, Sustain; British spelling per WRITING_SPEC.md rule 9). Kaizen is continuous incremental improvement involving everyone, standardised when changes work. — source: research sheet logistics-scm-frameworks.md, section 13.

No regional regulatory or reporting schemes (for example EU CSRD or ETS) are named or characterised; ESG is taught only as the brief's category definition. — source: research sheet logistics-scm-frameworks.md, section 12 guidance, and WRITING_SPEC.md rule 5.2.

Referenced lessons by name only, without re-teaching: 1.3 The Seven Rights of Logistics; 1.4 Types of Logistics: Inbound, Outbound, Reverse, and Third-Party; 3.2 Modes of Transport: Sea, Air, Road, Rail, and Multimodal; 6.1 Commercial Invoice and Packing List; 7.2 Air Cargo Operations and Cargo Acceptance ("never sign a cargo description you cannot stand over"); 8.1 Warehouse Functions and Layout (travel waste); 8.4 Warehouse Documentation and Stock Reconciliation; 10.1 Components of Total Logistics Cost; 11.3 Supply Chain Security and Cargo Integrity; 11.4 Managing Shipment Delays, Damage, and Discrepancies (root cause analysis); 12.2 Transit Logistics and Border Operations (border formalities worked early, never worked around); 12.3 Coordinating Carriers, Agents, and Regulatory Authorities; 13.2 Transport Route and Network Optimisation (backhaul and empty running); 13.3 Inventory Optimisation and Service Levels; 13.4 Logistics Performance Indicators and KPIs (baseline-and-trend discipline, no universal benchmarks); 15.1 Logistics Strategy and Service-Provider Selection; 15.2 Contract Management and Service-Level Agreements; 15.3 Business Continuity and Supply Chain Resilience. — source: curriculum lesson briefs and existing drafts.

The reuse-before-repair-before-recycling-before-disposal ordering of waste responses, the characterisation of waste as material paid for twice, and the transparency test (would this decision survive being written down and read aloud) are general business and professional-ethics knowledge, presented without attribution. — source: general business knowledge.

No emissions factors, no emissions percentages, and no percentage benchmarks of any kind are stated; all quantities, currency amounts and durations in the worked examples (4,000 and 1,500 US dollars per consignment; one week air and three weeks sea transit; 100, 40 and 62 pick errors; three quarters of volume by sea) are illustrative for teaching purposes. — source: WRITING_SPEC.md rule 5.1.

Module 15 Assessment — 12 questions

Enter via the instructor portal's Configure assessment step. Green ticks mark correct answers — do not enter them; enter options as-is.

Question 1 (tests lesson 15.1)

1

A national pharmaceutical wholesaler based in Accra supplies hospitals and clinics across the country from its own cold-chain-capable warehouse. Volumes are large and steady throughout the year, the delivery operation is tightly regulated and customer-critical — hospitals depend on daily deliveries — and the company's cold-chain handling and delivery reliability are widely regarded as the reason customers choose it. Capital is available. The finance director proposes outsourcing the entire delivery operation to a general transport provider quoting rates well below the operation's current cost, arguing that transport is never a core activity.

Which option best describes the sound make-or-buy analysis of this proposal?

1. The outsourcing should proceed, because capability, volume, control and capital are secondary considerations next to the quotation: a provider offering any complete transport service at a lower rate has, by that fact alone, proved itself the better source.

2. The four factors must each be answered honestly, and here they point against full outsourcing: the company already possesses specialist cold-chain capability; its volumes are large and steady enough to keep owned resources busy; the flow is customer-critical and tightly regulated, which argues for keeping it close or holding control through strict performance measures; and available capital weakens the saving argument — so the sounder position is to retain the activity, or outsource only parts of it under measurable commitments. ✓

3. The outsourcing should proceed, because a service contract transfers accountability for delivery performance to the provider, and the wholesaler would no longer need to monitor deliveries or retain internal knowledge of the operation.

4. The decision should turn on the control factor alone: because the flow is regulated, no logistics activity of any kind may ever be outsourced by any business, whatever the other factors show.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The make-or-buy decision is walked through four factors, and rarely does one decide alone. Capability: the wholesaler already performs specialist cold-chain distribution — trained staff, temperature-controlled storage, a delivery record customers buy — so the market is not obviously better here, and the general provider quoting less is pricing a different, simpler service. Volume: large, steady demand keeps owned vehicles and warehouses busy, which is exactly the condition under which ownership earns its keep; it is idle capacity, not ownership itself, that wastes capital. Control: the flow is customer-critical and tightly regulated, and the working rule says keep what you cannot safely surrender control of, or hold control through strict performance measures instead. Capital: with capital available, the saving argument loses the force it carries where ownership would starve the core business. Answer 1 is the price-first reflex: a lower quotation for a different service proves nothing until the services are comparable. Answer 3 repeats the most dangerous sentence in outsourcing — the task transfers, the accountability does not, and the wholesaler would remain answerable for every failed hospital delivery. Answer 4 commits the opposite shortcut: control argues for care, not for an absolute ban, because the discipline is to answer all four factors honestly and respect the result, hybrid positions included.

The takeaway: make-or-buy is a four-factor analysis performed activity by activity — and the lowest quotation is the beginning of the question, not the end of it.

Question 2 (tests lesson 15.1)

2

A cocoa exporter in Takoradi is selecting a provider for its European export consignments. Before any proposal was opened, the panel agreed a written evaluation framework: criteria drawn from the specification, weighted so that reliability, compliance and systems outrank price; a common scoring scale applied to every bidder; and documented reasoning behind every score. When the weighted totals are tabled, the cheapest bid ranks last of the short list on compliance and systems. The finance director proposes re-weighting the criteria so that price outranks the rest, arguing that the original weights were only a draft and the cheapest bid must be the safe choice.

Which option best assesses the proposal?

1. The proposal is sound, because weights are only arithmetic and changing them after the scores are known simply corrects a drafting error; the documentation can be updated to match the new totals.

2. The proposal is sound, because price is the only criterion that can be verified objectively, so it must always carry the greatest weight in any evaluation framework.

3. The scoring should be abandoned and the award decided on the reference checks alone, since what customers say is more reliable than any arithmetic.

4. The proposal should be rejected: the weights were agreed before any proposal was opened precisely to keep the evaluation honest, defensible and immune to pressure, and price is considered last — weighed against a genuinely comparable service and its total cost of ownership over the contract term, not against the other bidders' quotations alone. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

The evaluation framework's whole purpose is to be agreed before any proposal is opened. Criteria drawn from the specification, weights reflecting the service promise, a common scoring scale and documented reasoning are what keep the decision honest under pressure, let the losing bidder be told fairly why they lost, and protect the business if the award is later questioned. Re-weighting after the scores are known reverses that logic: the arithmetic stops being a measurement and becomes a justification for a decision already taken on price. Price, moreover, is considered last by design — meaningful only once the service being priced is genuinely comparable — and the cheapest bid here is the weakest on compliance and systems, exactly the criteria whose failures land on the customer's own shipments. Total cost of ownership asks what that record would cost in claims, expediting and management attention over the term, costs that never appear on a rate card. Answer 1 treats the weights as disposable arithmetic, which is the corruption itself; a framework re-cut to reach a preferred answer was never a framework. Answer 2 is the price-first fallacy: price is no more objective than the evidence behind a reliability or compliance score, and weighting it first recreates the award-on-the-only-number-available failure. Answer 3 discards a documented, comparable process for an impression; reference checks are a verification step inside the framework, not a substitute for it.

The takeaway: weights agreed in advance are the evaluation's integrity — and price is judged last, on total cost over the term.

Question 3 (tests lesson 15.1)

3

A manufacturer of bottled beverages in Kumasi contracted a third-party logistics provider to run its deliveries to retail distributors. For weeks the provider's vehicles have arrived late and short, and a key retail chain's buyer now telephones the manufacturer to say its stores are being left unstocked, warning that orders will move to a competitor. The logistics manager proposes replying that the delays are entirely the contracted provider's failure, that the buyer should raise them with the provider directly, and that the manufacturer holds no responsibility for work it no longer performs.

Which option best assesses the response?

1. The response is wrong: outsourcing transfers the task, not the accountability — the end customer experiences the failure as the manufacturer's own, so the manufacturer must manage the provider actively through the agreed measures and remedies, repair the customer relationship, and retain enough internal knowledge to specify, judge and, if necessary, replace the provider. ✓

2. The response is correct: a service contract allocates the customer experience to the provider, so the retail chain's only proper course is to pursue the provider under the provider's own contract.

3. The response is wrong, but for the opposite reason: ownership is the only real form of control, so the manufacturer must cancel the contract and rebuild its own fleet at once, before any customer conversation takes place.

4. The response is correct, provided the manufacturer also withholds the provider's entire invoice for the period, since improvised penalties are the remedy the contract intended for misses of this kind.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

Outsourcing transfers the task, not the accountability. The retail chain's buyer does not contract the provider and does not care who drives the trucks; it experiences the failure as the manufacturer's failure and takes its orders elsewhere — which is why the first response is to establish what happened, hold the provider to the agreed remedies, and repair the only relationship the end customer can see. Managing the provider actively — through the specification, the KPIs, the reporting and the review rhythm — is precisely what the retained accountability requires, together with enough internal knowledge kept inside the business to specify the service, judge the performance and, when necessary, leave the relationship; that knowledge is also the antidote to the dependency that long single-provider operation quietly builds. Answer 2 states the contract-transfer misconception outright: no clause moves the customer's experience to the provider, and pointing the buyer at the 3PL simply confirms that nobody is managing the promise. Answer 3 reaches the opposite conclusion by the wrong route — ownership is only one form of control, and a fleet rebuilt in a crisis would arrive late for the same buyers; the failure indicates weak provider governance, not the absence of painted doors. Answer 4 improvises a penalty the contract did not write; remedies are applied as agreed, and withholding payment outside the agreed machinery teaches both sides that the document is decorative.

The takeaway: outsourcing done well is delegation with control — pointing the customer at the provider is abdication with paperwork.

Question 4 (tests lesson 15.2)

4

A distributor based in Cape Coast receives a draft service-level agreement containing the commitment: "The provider will achieve an on-time in-full delivery rate of 95 per cent every month." The contract manager declares the commitment now fully measurable and enforceable, because it contains a number and a target. All figures in this question are illustrative for teaching purposes.

Which option best assesses that judgement?

1. The judgement is correct: a stated percentage is the definition of a measurable commitment, and no further drafting is required once a target appears in the document.

2. The commitment should be strengthened by adding more indicators — a dozen or more KPIs covering every aspect of the service — because a contract carrying many measures cannot be disputed.

3. The judgement is premature: a number alone is not yet a commitment — the clause still needs the indicator's formula, under which an order counts only if delivered both on time and in full; the counting window with rule-bound exclusions; the defined data source and measurer; the reporting cadence; and the pre-agreed remedy that applies when the target is missed. Without the source, every miss becomes an argument about arithmetic; without a remedy, the target is a wish. ✓

4. The judgement is premature because percentages are never enforceable; the clause should be replaced by a general duty to deliver promptly, drafted in legally precise language.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

A number is necessary but nowhere near sufficient. The drafting discipline is that two strangers, given the same data, would score the commitment identically — and this clause fails that test four ways. Its formula is unstated: on-time in-full counts an order only if it arrives both on time and in full, and that strictness must be written down. Its counting window, its exclusions and its reporting cadence are absent. Its data source and measurer — which system of record supplies the numbers, and who computes the score — are absent, and this is the defect that turns every missed month into a dispute: one side's system says the target was met, the other's says it was missed, and the difference is the argument. And no remedy is attached, so a miss changes

nobody's behaviour: an SLA without consequences is a wish list. Answer 1 accepts the number as the finish line, mistaking a target for a commitment. Answer 2 reaches for quantity of measurement, but few measures, well chosen, outperform a long list nobody reads — a contract carrying dozens of KPIs is as blind as one carrying none. Answer 4 moves backwards: legal polish cannot make an unmeasurable promise countable, and a general duty to deliver promptly is exactly the aspiration this clause was meant to replace.

The takeaway: the difference between a promise and a commitment is a number, a method, a source and a consequence — all four, or it is none of them.

Question 5 (tests lesson 15.2)

5

A furniture importer in Accra holds a warehousing and delivery contract whose service-level agreement sets an on-time in-full target, a service-credit schedule that applies only after consecutive months below target, and an escalation ladder: review at the operational meeting, corrective-action plan, service credits, and an exit trigger reserved for persistent failure with no effective plan. This month, for the first time since the contract began, the on-time in-full result falls below target. The commercial manager proposes immediately withholding the whole month's payment and threatening termination, to send the provider a message.

Which option best describes the correct response under the agreement?

1. The proposal is sound: a first miss should be answered with the strongest available consequence, because severity at the start prevents every later failure.

2. The proposal bypasses the machinery both parties agreed: a first miss is review territory under the ladder — the result is tabled at the operational meeting, the provider's explanation is logged, and a corrective action is agreed and tracked to closure — while the credits wait for the consecutive-miss threshold and the exit trigger waits for sustained, uncorrected failure; improvising harsher measures teaches both sides that the document is decorative. ✓

3. The miss should simply be ignored, because service-level targets are statistical averages that carry no consequence unless they are missed repeatedly for a whole year.

4. The target should be renegotiated downwards at once, so that the provider is not discouraged and the relationship is preserved while performance is restored.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The ladder's value lies in being followed as written, and that discipline condemns improvisation in both directions. A first miss is the ladder's first rung: the result appears in the agreed report, is tabled at the operational meeting, the provider explains, the miss is logged, and a corrective action is agreed and tracked. Many misses end there, which is the design. The service credits wait for the threshold the parties wrote — consecutive months below target — and the exit trigger waits for sustained failure with no effective corrective plan; this scenario meets neither condition. Consequences were priced in advance precisely so that neither mood nor discretion sets them, and the customer who shouts in month one teaches the provider that the document can be renegotiated by volume. Answer 1 is escalation by temper: severity is not a substitute for the procedure, and the strongest consequence deployed first leaves nothing in reserve and no log of what actually happened. Answer 3 errs the other way: a miss on a measured commitment is never mere information — even at step one it must be made visible, explained and corrected, and the issue log exists so that a first miss and a pattern are told apart. Answer 4 rewrites the deal mid-failure; targets are set from evidence just beyond the baseline, and lowering one because it bit once destroys the discipline for every month after.

The takeaway: an SLA is proven in the month the target is missed — the ladder run as written turns failure into procedure instead of conflict.

Question 6 (tests lesson 15.2)

6

A wholesale business in Tamale receives its provider's monthly report: the on-time in-full and order-accuracy targets have both been missed. The managing director declares the provider in breach of contract, instructs that the dispute-resolution clause be invoked, and asks lawyers to prepare termination letters — treating the missed scores as a legal rupture. The record shows the misses are recent, were explained at the operational meeting, and are covered by corrective-action plans with recovery already visible.

Which option best assesses the managing director's position?

1. The position is correct: any result below a target stated in the service-level agreement is by definition a breach of a contractual obligation, and escalation to legal mechanisms is the required next step.

2. The position is correct in substance but premature in timing: the lawyers should be engaged and the termination letters prepared the moment the second month's scores arrive, whatever the corrective-action plans show.

3. The position is wrong because service-level targets are voluntary aspirations with no contractual force, so no consequence of any kind can ever follow a missed target.

4. The position confuses two things the contract keeps apart: a missed performance standard is a measured service failure answered through the SLA's machinery — review,

corrective-action plan, service credits — while escalation and the dispute provisions are reserved for failures of obligation and for persistent failure that survives correction; treating ordinary misses as legal ruptures destroys the ladder's purpose, which is to keep failures management matters. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

The distinction between obligations and performance standards runs in both directions. A contractual obligation is a duty to do or not do something — maintain defined insurance, submit the agreed reports, handle goods with care — established by documents and records, and answered through escalation and, if it persists, the dispute and termination provisions. A performance standard is a graduated level to achieve, established by measurement against a defined method, and answered through the SLA's machinery: review, corrective action, service credits. The scenario describes measured misses — recent, explained, with corrective plans and visible recovery — which is ladder territory, not legal rupture. The ladder exists precisely so that ordinary failures stay management matters; escalating to lawyers at the first red month would dismantle the machinery both parties agreed to fund and operate. Answer 1 collapses the standard into the obligation and reads the contract as a trap that springs on any miss. Answer 2 only relocates the trap: termination is governed by the contract's own definition of cause — persistent failure that survives correction — not by a count of red months alone. Answer 3 overcorrects into the opposite defect: targets with no consequences are a wish list, and calling them unenforceable removes exactly the pressure that drives corrective action. One caution completes the picture: the compliance review should also confirm that no duty — insurance, reporting, confidentiality — has failed quietly alongside the red scores, because the two can move in opposite directions.

The takeaway: a missed target is not automatically a broken obligation — and a broken obligation is never measured by the scorecard alone.

Question 7 (tests lesson 15.3)

7

A fruit-processing exporter in Tamale depends on a single overseas supplier for its packaging materials and moves finished goods to port along one road corridor that floods in the rainy season. Management approves four actions: servicing the vehicles and clearing drainage along the corridor before the rains; qualifying a second packaging supplier and documenting the alternative corridor with its costs, timings and contacts; paying the annual insurance premium under which flood damage to finished stock would be compensated; and, after one flood, pumping out the depot and restoring the damaged section of the route to service.

Which option correctly classifies the four actions?

1. Vehicle servicing and drainage clearing are risk prevention; qualifying the second supplier and documenting the alternative corridor are contingency planning; the insurance premium is risk transfer; pumping out the depot and restoring the route afterwards is disaster recovery. ✓

2. Vehicle servicing and drainage clearing are contingency planning; qualifying the second supplier and documenting the alternative corridor are risk prevention; the insurance premium is contingency planning; pumping out the depot and restoring the route afterwards is risk prevention.

3. All four actions are risk prevention, because each was arranged before or shortly after the event with the aim of reducing the company's losses.

4. The insurance premium is disaster recovery, because the compensation restores the money after the loss; the other three actions are contingency planning, because all of them prepare alternatives.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

The three operational layers answer different questions on different sides of the event, and the scenario supplies one action for each, plus risk transfer. Servicing vehicles and clearing drainage act before the event to make it less likely or less damaging: risk prevention. Qualifying a second packaging supplier and documenting the alternative corridor assume the event will happen anyway and prepare the ready answer: contingency planning, whose product is not a reduced likelihood but a usable fallback — the qualified alternative, the documented route, priced and contactable in advance. Pumping out the depot and restoring the flooded route after the event is the after-discipline: disaster recovery, the restoration of facilities and operations. The insurance premium is none of the three operational layers: it is risk transfer, compensating the financial loss afterwards while keeping nothing moving, serving no customer and running no system by itself. Answer 2 swaps the two before-the-event layers — the drainage work reduces the likelihood of flooding but prepares no alternative — and files risk transfer under contingency, mistaking a policy for a plan. Answer 3 erases the distinctions the lesson drew, and with them the discipline of asking of every arrangement: does this shrink the odds, shorten the fall, or set the pace of getting up? A business strong in only one layer is prepared for only one half of a disruption. Answer 4 mistakes compensation for restoration: the money returns, but the goods, the customers and the operation do not run on a policy.

The takeaway: prevention changes the odds, contingency prepares the alternative, recovery restores afterwards, and insurance pays — four different purchases a continuity programme buys deliberately.

Question 8 (tests lesson 15.3)

8

The order-processing system of a wholesale business in Kumasi fails. Before the event, management had set two recovery objectives for the system in writing: the system must be restored and working again within one working day, and it may lose no more than the last hour of recorded transactions. The recovery arrangements, however, run a single backup of the system once every night.

Which option correctly identifies the two objectives and assesses the backup arrangement?

1. The one-day restoration target is the recovery point objective and the one-hour data-loss limit is the recovery time objective; the nightly backup satisfies both, because a backup made every night is by definition an hourly one.

2. Both objectives are recovery time objectives, because both measure the speed with which the system returns to service; the backup frequency is an operational detail with no connection to either objective.

3. The one-day restoration target is the recovery time objective and the one-hour data-loss limit is the recovery point objective — and the nightly backup cannot honour them: a failure late in the working day puts many hours of transactions at risk, far beyond the one-hour limit, so the backup regime and the recovery arrangements must be redesigned to meet the objectives and then verified by testing. ✓

4. The two objectives were set wrongly, because recovery objectives can only be defined after a failure has shown what is realistically achievable; the nightly backup should continue untouched until then.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The two objectives measure different clocks. The recovery time objective is the targeted maximum period within which the disrupted system should be restored — the one-working-day figure asks how fast the system returns. The recovery point objective is the maximum acceptable amount of data loss measured in time before the disruption — the one-hour figure asks how much of the system's information survives. The nightly backup fails the pairing: a failure late in the working day leaves the last backup at the previous midnight, so the

transactions at risk run to many hours, far beyond the limit management set. The objectives come first, agreed before any event; the backup schedule and the recovery arrangements are then designed to meet them, and readiness is proven in test — a failover test measures the achieved restoration time against the recovery time objective, and the backup regime against the recovery point objective. Answer 1 swaps the two definitions and then declares the backup adequate by a definition no lesson teaches. Answer 2 collapses the pair into speed and severs backup frequency from the objectives, when backup frequency is precisely what sets how much data a failure can take between records. Answer 4 inverts the discipline: objectives improvised after a failure are a description of damage, not a commitment — and a business that first meets its disruption on the day it happens is not responding, it is discovering.

The takeaway: the recovery time objective is the clock on restoration, the recovery point objective is the clock on data — both set in writing before the event, both proven in test.

Question 9 (tests lesson 15.3)

9

An importer of machine parts in Kumasi is reassured by its logistics coordinator: "We are not concentrated — our containers move under three different shipping lines, cleared by two different agents." In fact, all three lines route via the same overseas corridor and discharge at the same port terminal, both agents submit declarations through the same electronic clearance system, and one senior clerk holds the only working knowledge of the transit documentation.

Which option best assesses the reassurance?

1. The reassurance is correct: dependence is counted by the number of contracts signed, and three shipping lines with two agents is a diversified arrangement that needs no further review.

2. The reassurance is wrong, and the remedy is to eliminate every concentration immediately, whatever the cost, because any concentration at all is a mistake no professional operation may carry.

3. The reassurance is wrong only about the clerk: people can always be replaced at short notice, so they are never single points of failure; the corridor, the terminal and the system are safe because no failure has yet occurred there.

4. The reassurance counts diversification at the wrong level: the concentration map traces each critical activity's dependencies down to the physical chain, so the single corridor, the single terminal, the single clearance system and the clerk's unique knowledge are the true single points of failure — each to be circled, then covered by mitigations whose standing cost is accepted the way an insurance premium is accepted, with the map redrawn annually as the business changes. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

Dependence is counted at the level of the physical chain, not the number of contracts. The concentration map traces each critical activity's dependencies and circles the ones with no alternative: if all three shipping lines ride the same corridor into the same terminal, a single closure or terminal failure stops every container regardless of whose name is on the bill of lading; if both agents submit through one clearance system, the system is a single point of failure; and people count as much as ports — the clerk's unique knowledge is among the most fragile dependencies on the map. The lesson's warning fits exactly: a business can hold contracts with three carriers and still be fatally concentrated. Answer 1 counts diversification at the wrong level, which is how concentration stays invisible — its savings accrue daily in the cost reports while the exposure is charged in full on one afternoon. Answer 2 over-corrects: concentration is not automatically a mistake, because redundancy costs money and sometimes no alternative exists; the discipline is to name and price the exposure, then buy mitigations deliberately where the impact would concentrate — a qualified alternative corridor, a second clearance route, cross-trained staff, a buffer on the critical lines — accepted the way an insurance premium is accepted, against a rare severe loss. Answer 3 is wrong twice: people are prime single points of failure, and no failure yet is not evidence of no risk — which is why the audit is an annual duty, not a one-off project.

The takeaway: a concentration you have mapped, priced and covered is a managed risk; one you have never drawn is a surprise with your name on it.

Question 10 (tests lesson 15.4)

10

A beverages exporter based in Tema has moved the steady, forecastable replenishment base of its European flow from air to sea, keeping air freight for genuinely urgent consignments. The marketing team now asks the logistics manager to publish the exact percentage by which the change reduced freight emissions, and to adopt a European retailer's published emissions figures as the company's new targets, since both would make a stronger claim.

Which option best describes the professionally sound response?

1. Both requests should be accepted: the mode ranking is precise enough to yield exact percentage savings, and figures published by other organisations are the most defensible possible targets.

2. Both requests should be declined: the directional ranking — sea the lowest-emitting main mode per tonne-kilometre, then rail, then road, with air by far the highest — states

direction only, because the size of the gaps varies with the vessel, the vehicle, the load carried, the fuel and the method of measurement, so no honest exact figure exists; the company measures its own indicators, such as fuel per tonne-kilometre and the share of tonne-kilometres moved by each mode, and trends them against its own baseline, declining to import numbers produced under other conditions. ✓

3. The publication request should be declined, but for a different reason: no claim of any kind can be made, because the directional ranking of the modes is not reliable knowledge and the change may equally have increased emissions.

4. Both requests should be accepted in part: the percentage can be published provided the company counts only the emissions of vehicles it owns itself, since emissions created by purchased sea freight belong to nobody.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The directional ranking is one of the few environmental claims logistics can state with confidence: per tonne-kilometre, sea is the lowest-emitting of the four main modes, rail is next, road is higher still, and air is by far the highest — so shifting the forecastable base flow from air to sea turns the emissions trend in the right direction, and usually cuts cost in the same stroke. What the ranking does not yield is an exact percentage: the gaps vary with the vessel, the vehicle, the load carried, the fuel and the method of measurement, so any precise figure the manager published would be a decorated number rather than a measurement. The honest instrument is the operation's own trend: fuel per tonne-kilometre, the share of tonne-kilometres moved by each mode, tracked against the company's own baseline in the standard indicator format — a fleet whose fuel per tonne-kilometre falls year on year is measurably improving even if nobody can state its exact tonnage of carbon dioxide equivalents. Answer 1 accepts both decorated claims. Answer 3 denies the one confident direction the lesson teaches; the ordering follows the fuel burned to move a tonne, and its direction does not vary. Answer 4 is the outsourcing self-deception the three scopes dismantle: emissions contracted to a shipping line do not vanish — they move to Scope 3, and a serious footprint counts the freight it buys, not only the vehicles it owns.

The takeaway: measure what you can, trend it honestly, use the directional ranking to point the trend — and decline the number nobody can honestly compute.

Question 11 (tests lesson 15.4)

11

At a port inspection, an official tells a clearing agent that a consignment of imported cosmetics will be held for a full documentation examination — unless an informal payment is made, after which the consignment will "pass as presented". The agent notices that the packing-list quantities do not match the declared carton count, a genuine discrepancy that should be corrected before release. The company's code of conduct prohibits bribes and facilitation payments.

Which option best describes the situation and the professional response?

1. The payment demanded would be a bribe, not a facilitation payment: it is sought to change the outcome of the inspection and pass goods that should be examined and corrected — and even a genuine facilitation payment would breach the code, which prohibits both; the response is to decline courteously, escalate through the declared procedure, record the approach factually, and correct the document discrepancy through the proper channel. ✓

2. The payment would be an ordinary facilitation payment, because it is small, routine and common at the post, and it may properly be recorded in the trip expenses as a standard border cost.

3. The payment would be an ordinary facilitation payment, because it would only affect the timing of the release and not the decision itself, and timing alone is never a matter of integrity.

4. The agent should pay to protect the consignment and then say nothing internally, because reporting the approach would expose the company to scrutiny without changing what happened at the post.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

The distinction every professional must be able to state does the classifying here. A facilitation payment expedites a routine, non-discretionary government action whose outcome is already fixed — the documents are in order, the goods are as declared, and the only variable is speed. The payment offered in this scenario is not that: the consignment carries a genuine document discrepancy, the examination is what the process requires, and the money is sought to change the outcome — to pass goods that should be held and corrected. That is a bribe by definition. And even the narrower case would not help: the company code prohibits both bribes and facilitation payments, and the stricter standard that governs the professional is the code, because permitted somewhere is not the test. The response follows the professional sequence: decline courteously, escalate through the declared channel, record the approach factually — and then correct the packing-list discrepancy through the proper process, because the

underlying defect does not disappear with the refused payment. Answer 2 misclassifies and launders at once: booking the payment as a standard border cost misstates what happened, and a wrong payment is never repaired by a wrong record. Answer 3 applies the timing rationalisation to a case where timing is not what is on offer — the decision itself is for sale. Answer 4 hides the event entirely; escalation protects the professional as much as the organisation, and an unreported approach teaches the company nothing.

The takeaway: the test is what the payment is for — and ethics has no cost-benefit test, whatever the consignment and whatever the queue.

Question 12 (tests lesson 15.4)

12

The manager of a regional distribution fleet in Kumasi wants to reduce its fuel consumption and asks how to apply the plan-do-check-act cycle to the problem.

Which sequence best applies PDCA as the module teaches it?

1. Launch a fuel-awareness campaign across the fleet with posters and a launch event, review the total fuel bill after a year, and repeat the campaign if the figure disappoints.

2. Purchase more efficient vehicles immediately, because equipment is the only real cause of fuel consumption, and compare the new vehicles' consumption with the old fleet's when the first invoices arrive.

3. Measure fuel per tonne-kilometre to establish a baseline, apply root cause analysis to find the biggest underlying cause — for example, a high share of empty return legs — trial a backhaul arrangement on one lane, re-measure against the baseline after an agreed period, then standardise what worked and repeat the cycle on the next-biggest cause. ✓

4. Retrain all drivers once on general fuel-efficient driving, then measure total fuel consumption a single time at the end of the year to record the result of the training.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The cycle is a complete instruction. Plan from measurement: a baseline of fuel per tonne-kilometre, then root cause analysis to find the biggest underlying cause rather than the loudest symptom — here, empty return legs producing kilometres with no freight to show for them, the purest waste in transport. Do on a limited scale: a backhaul arrangement trialled on one lane, not imposed fleet-wide. Check against the baseline: re-measure after an agreed period and compare honestly. Act: standardise what worked into the operating routine, adjust what did

not, and restart the cycle on the next-biggest cause. Small, measured, repeated cycles compound; an empty leg fixed permanently saves fuel every week thereafter. Answer 1 is the campaign pattern the module's closing warning dismantles: broad, unmeasured against any baseline, and abandoned when the posters fade — no cause is ever found, so the waste resumes its daily work. Answer 2 buys equipment before measurement: new vehicles cannot fix empty running, because the waste lives in the network, not the engine, and without a baseline even the new fleet's performance cannot honestly be judged. Answer 4 trains everyone once and measures once: it assumes the cause is knowledge when the measured cause may be routing, skips the diagnosis entirely, and a single end-of-year figure provides no before-and-after comparison along the way — the symptom treated, the cause intact.

The takeaway: measure first, fix the cause, trial small, standardise what survives the check, and repeat — the rhythm matters more than the size of any single fix.